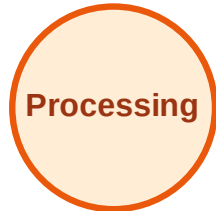


# BFS Traversal

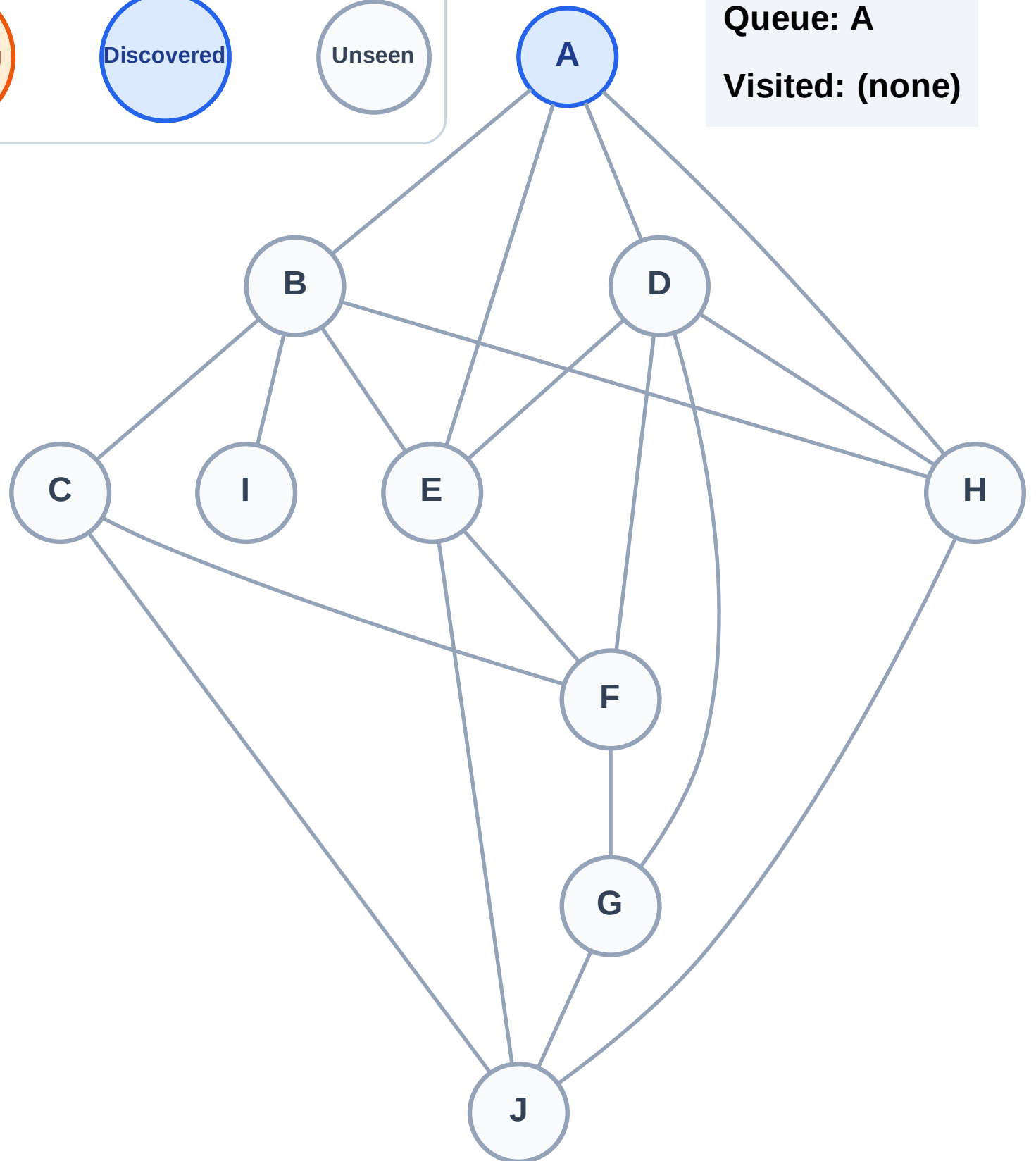
Step 1: Start at A

## Legend



Queue: A

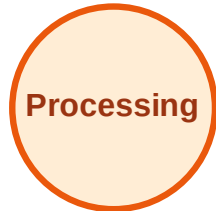
Visited: (none)



# BFS Traversal

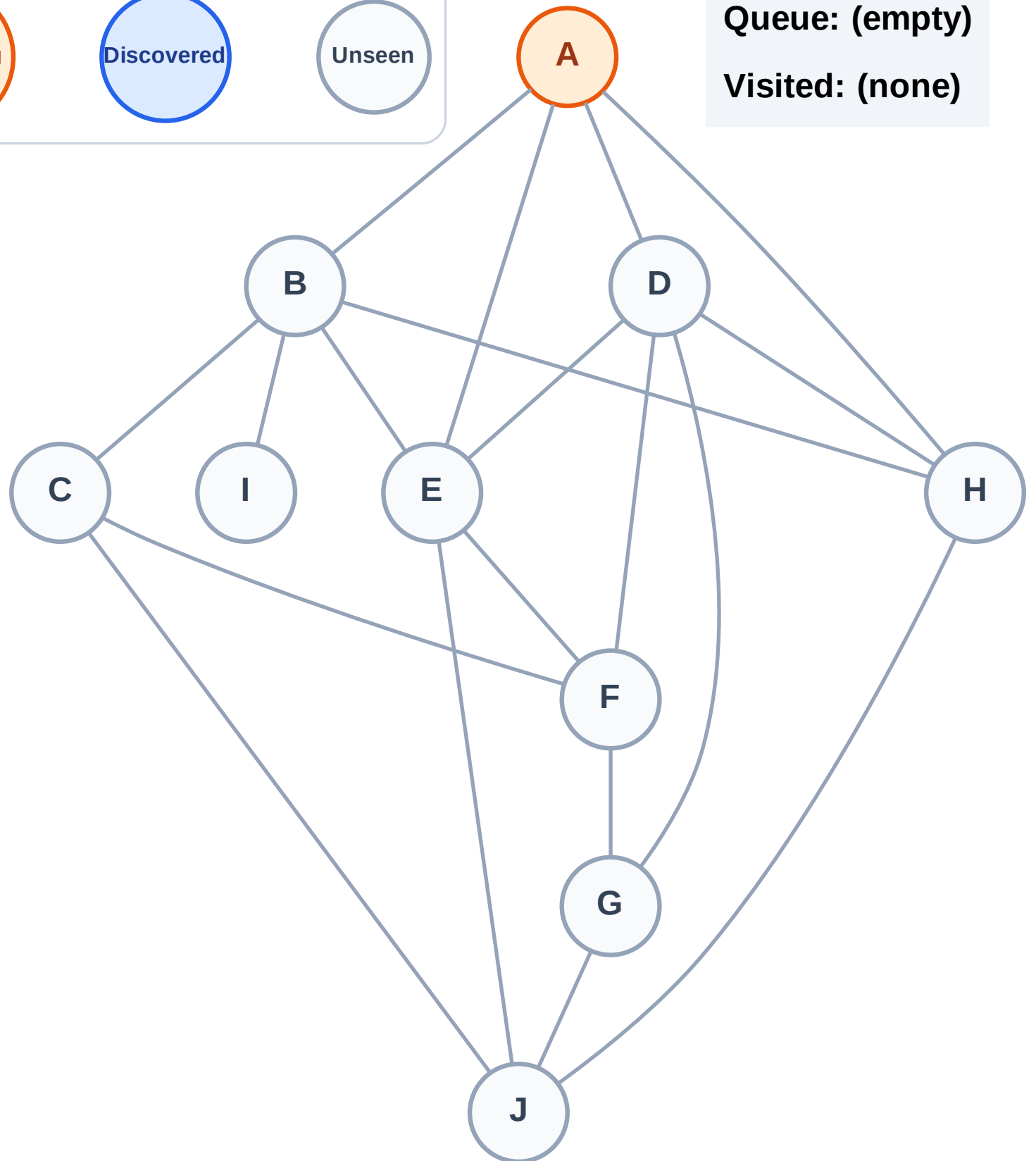
Step 2: Process node A

## Legend



Queue: (empty)

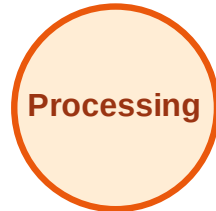
Visited: (none)



# BFS Traversal

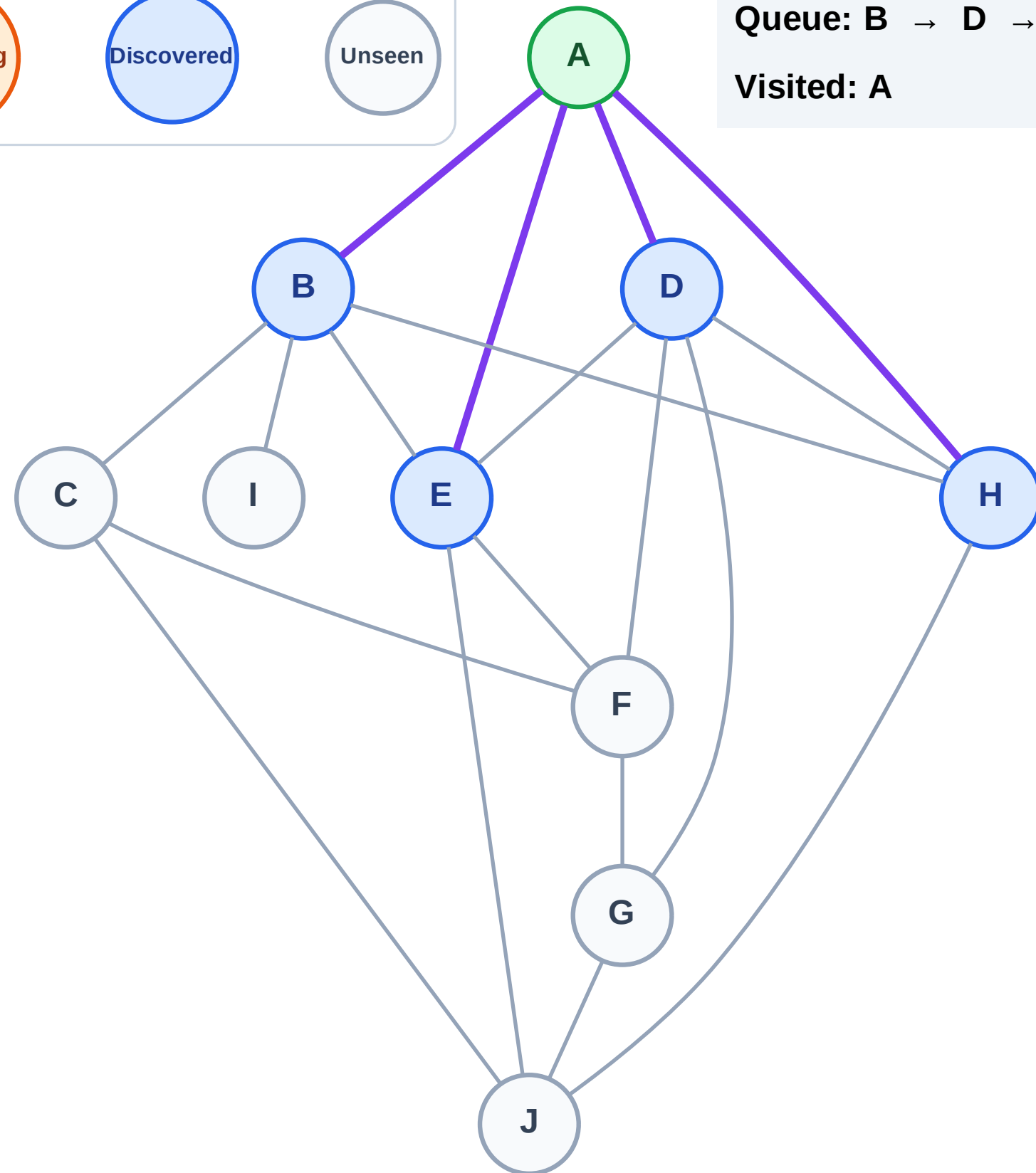
Step 3: Discover B, D, E, H from A

## Legend



Queue: B → D → E → H

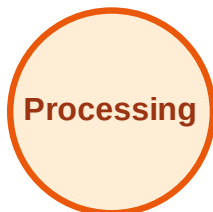
Visited: A



# BFS Traversal

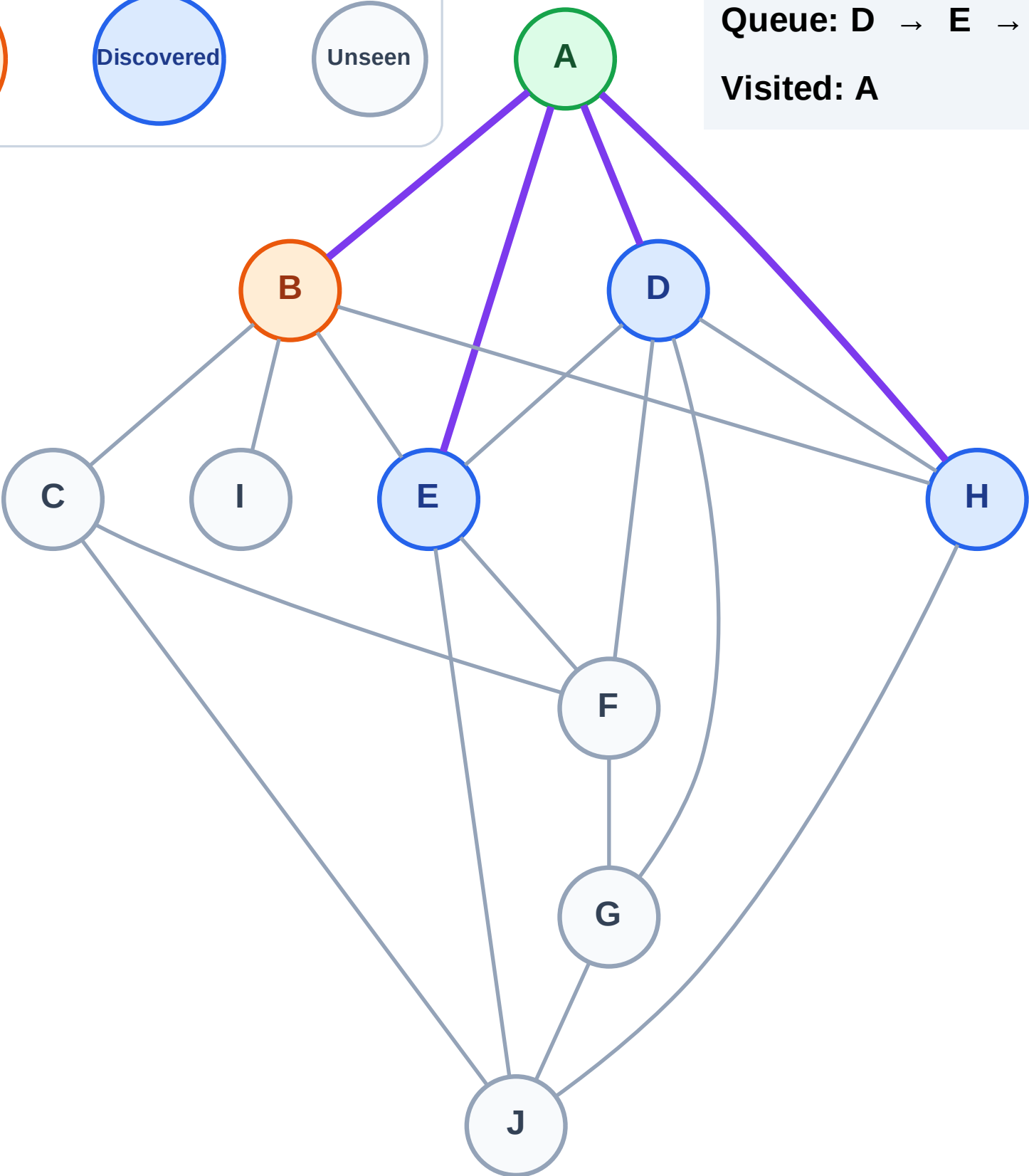
Step 4: Process node B

## Legend



Queue: D → E → H

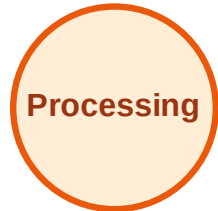
Visited: A



# BFS Traversal

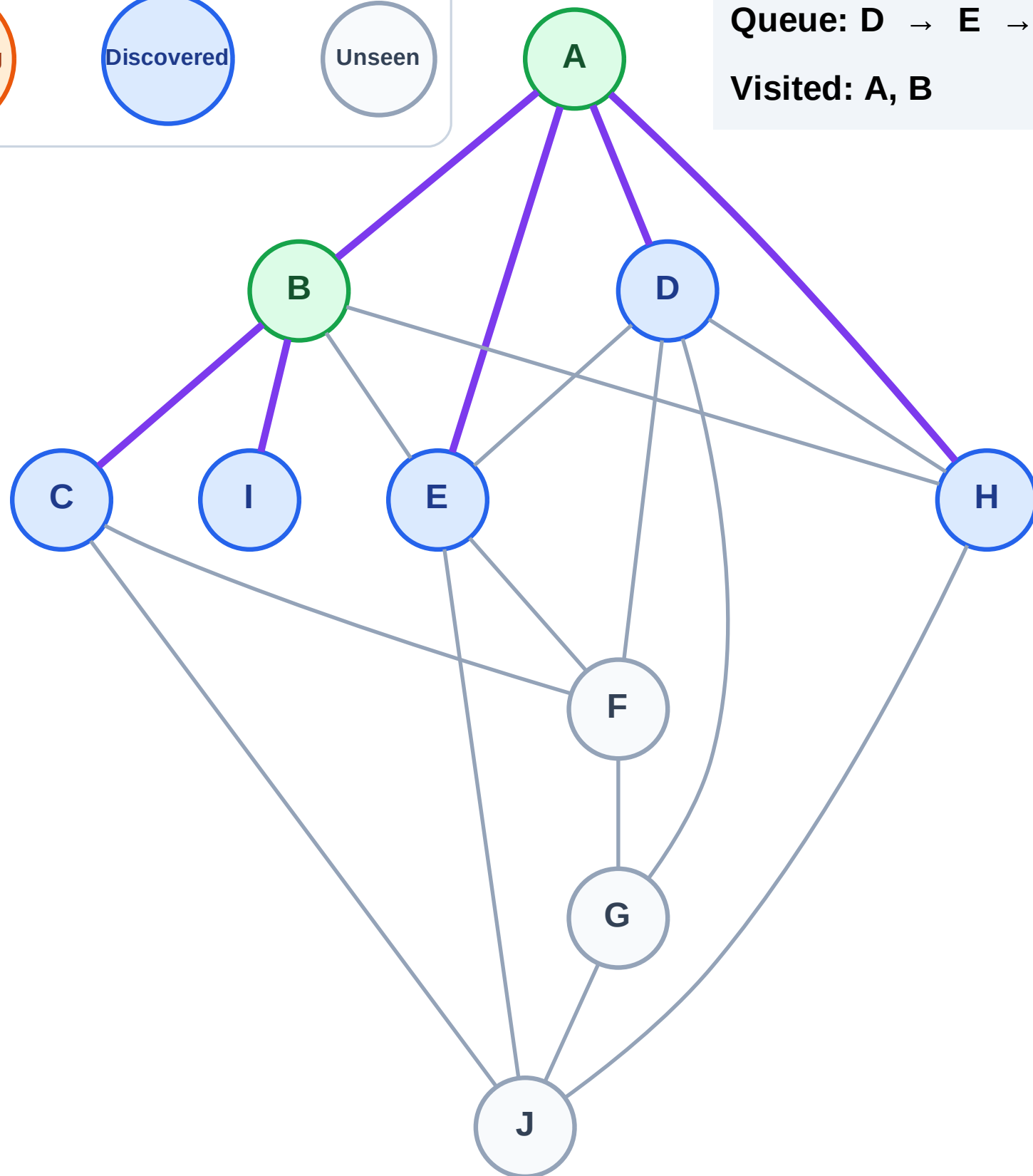
Step 5: Discover C, I from B

## Legend



Queue: D → E → H → C → I

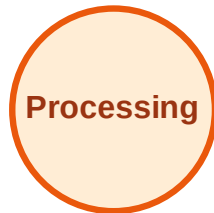
Visited: A, B



# BFS Traversal

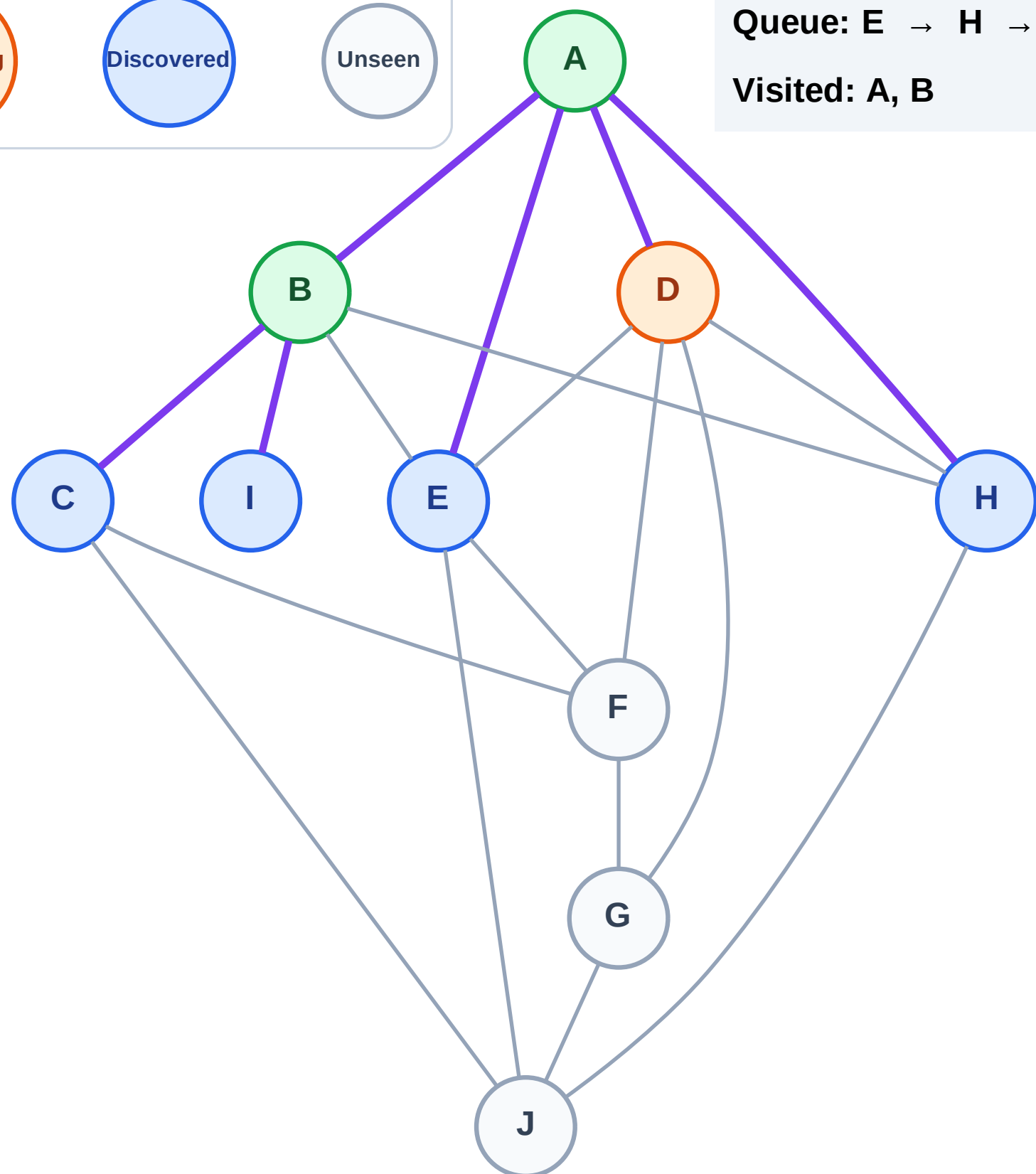
Step 6: Process node D

## Legend



Queue: E → H → C → I

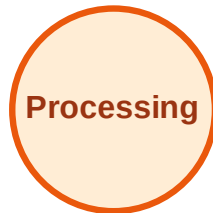
Visited: A, B



# BFS Traversal

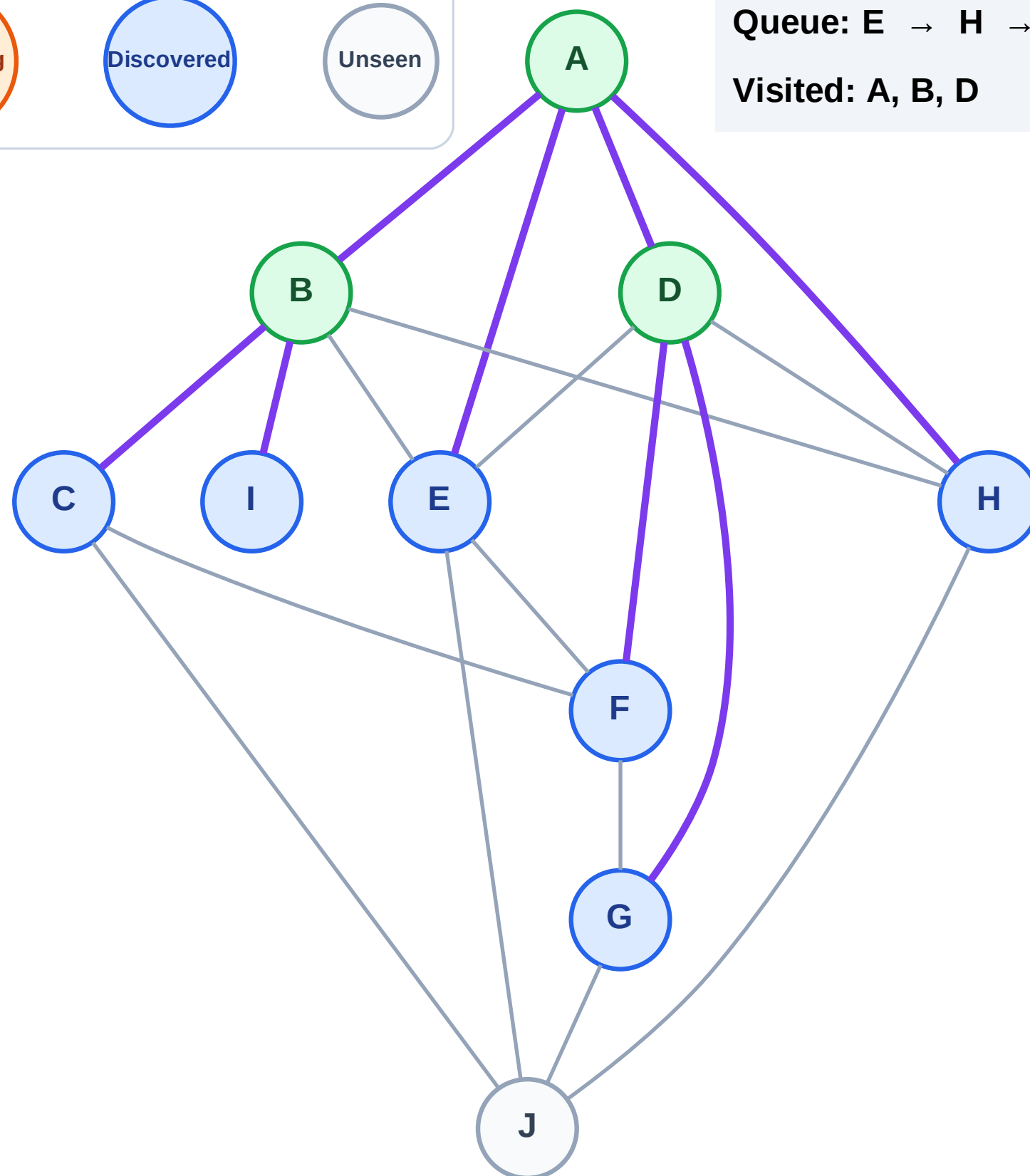
Step 7: Discover F, G from D

## Legend



Queue: E → H → C → I → F → G

Visited: A, B, D



# BFS Traversal

Step 8: Process node E

**Legend**

Complete

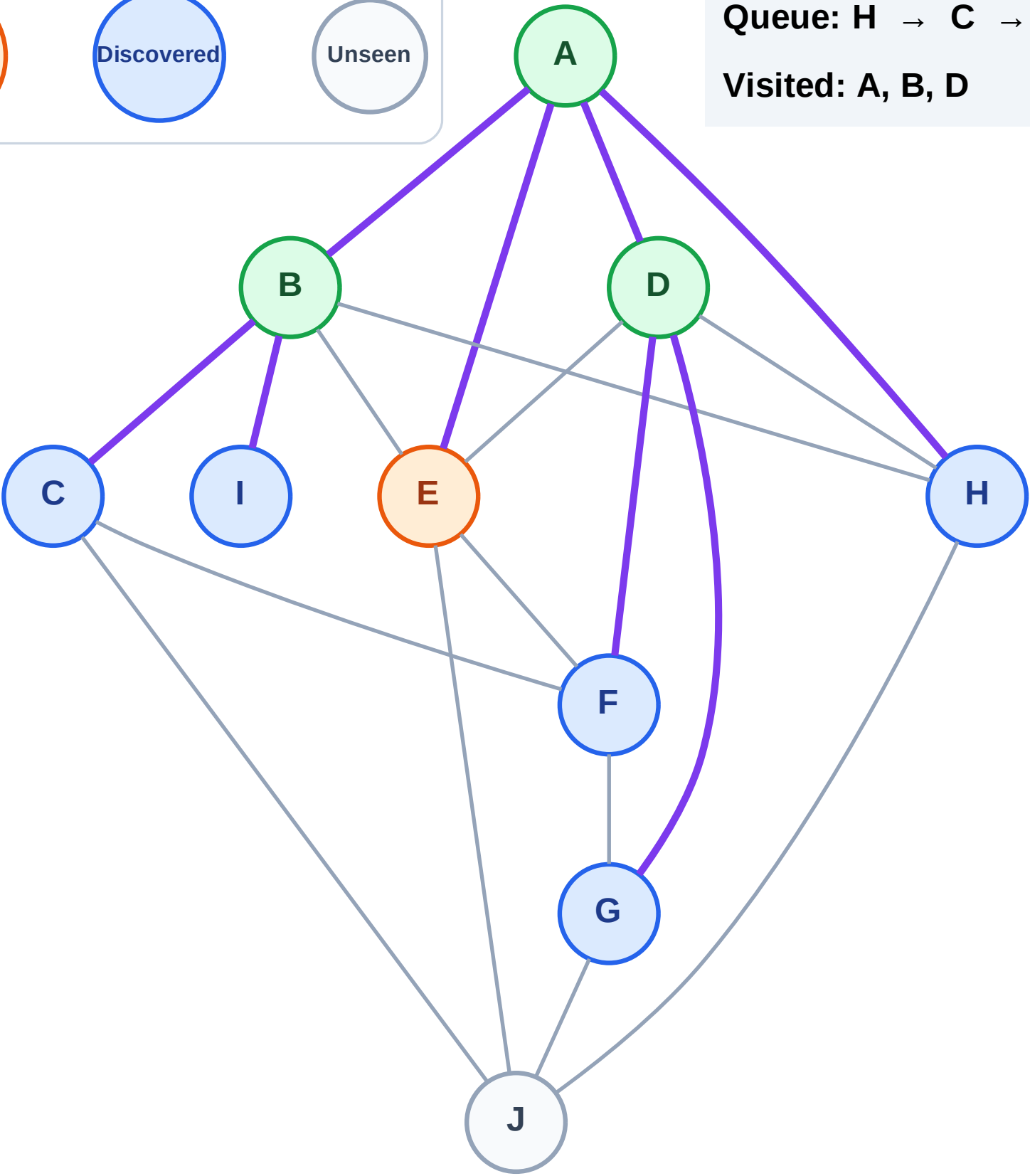
Processing

Discovered

Unseen

Queue: H → C → I → F → G

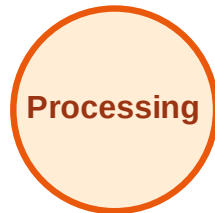
Visited: A, B, D



# BFS Traversal

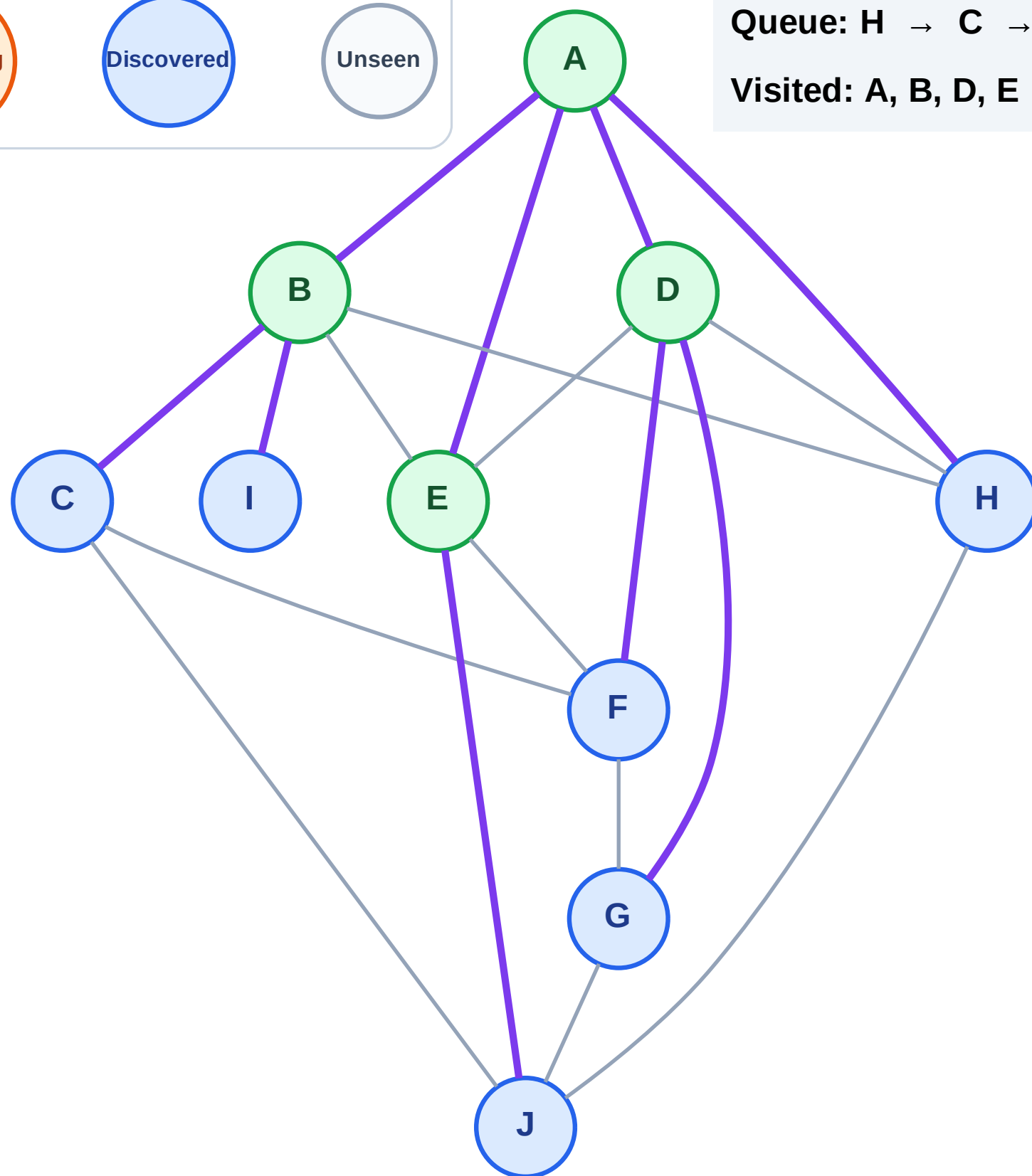
Step 9: Discover J from E

## Legend



Queue: H → C → I → F → G → J

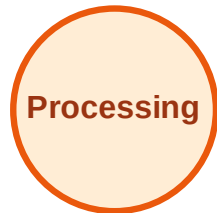
Visited: A, B, D, E



# BFS Traversal

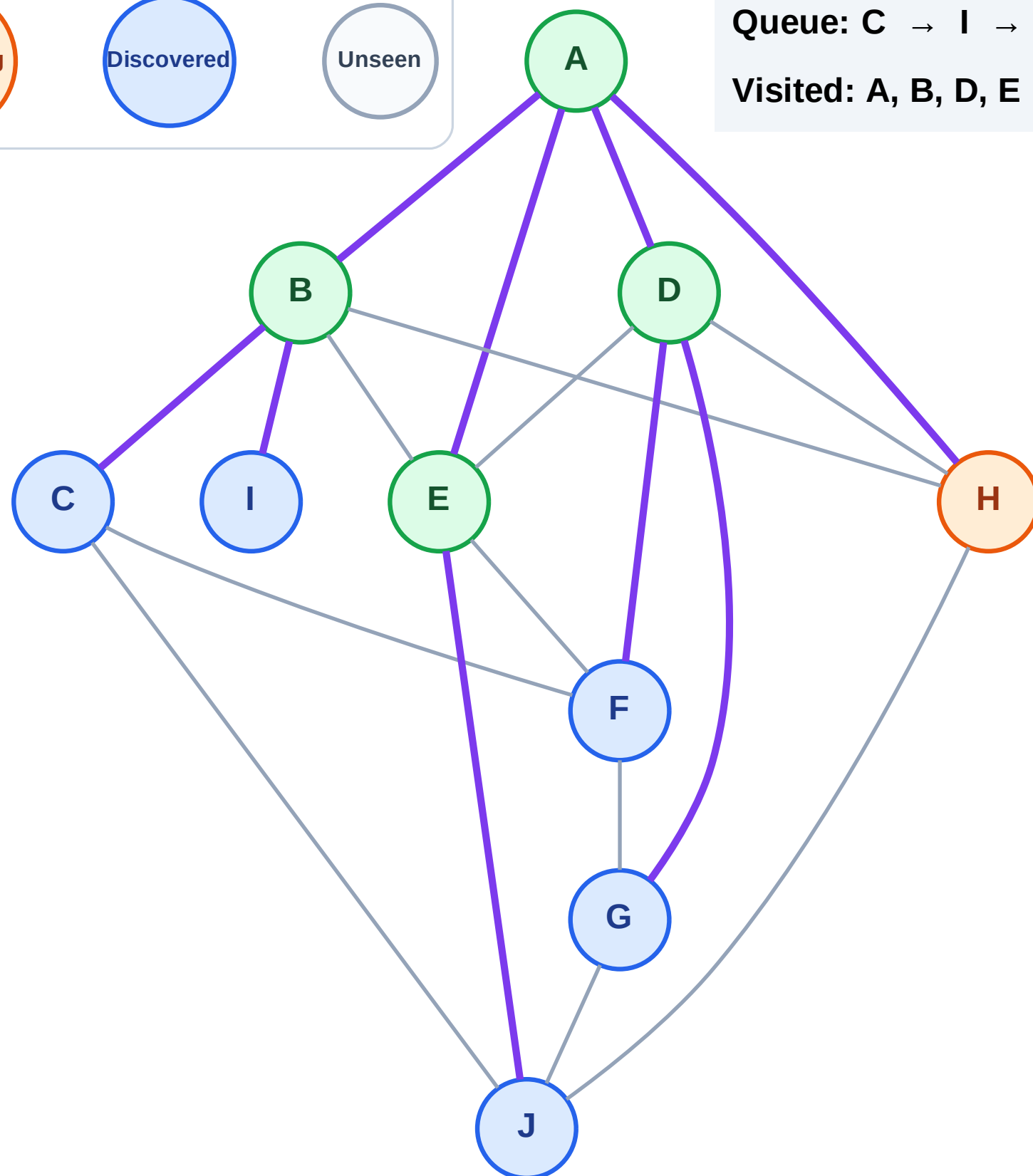
Step 10: Process node H

## Legend



Queue: C → I → F → G → J

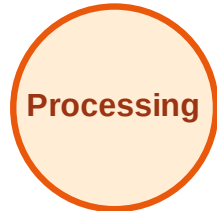
Visited: A, B, D, E



# BFS Traversal

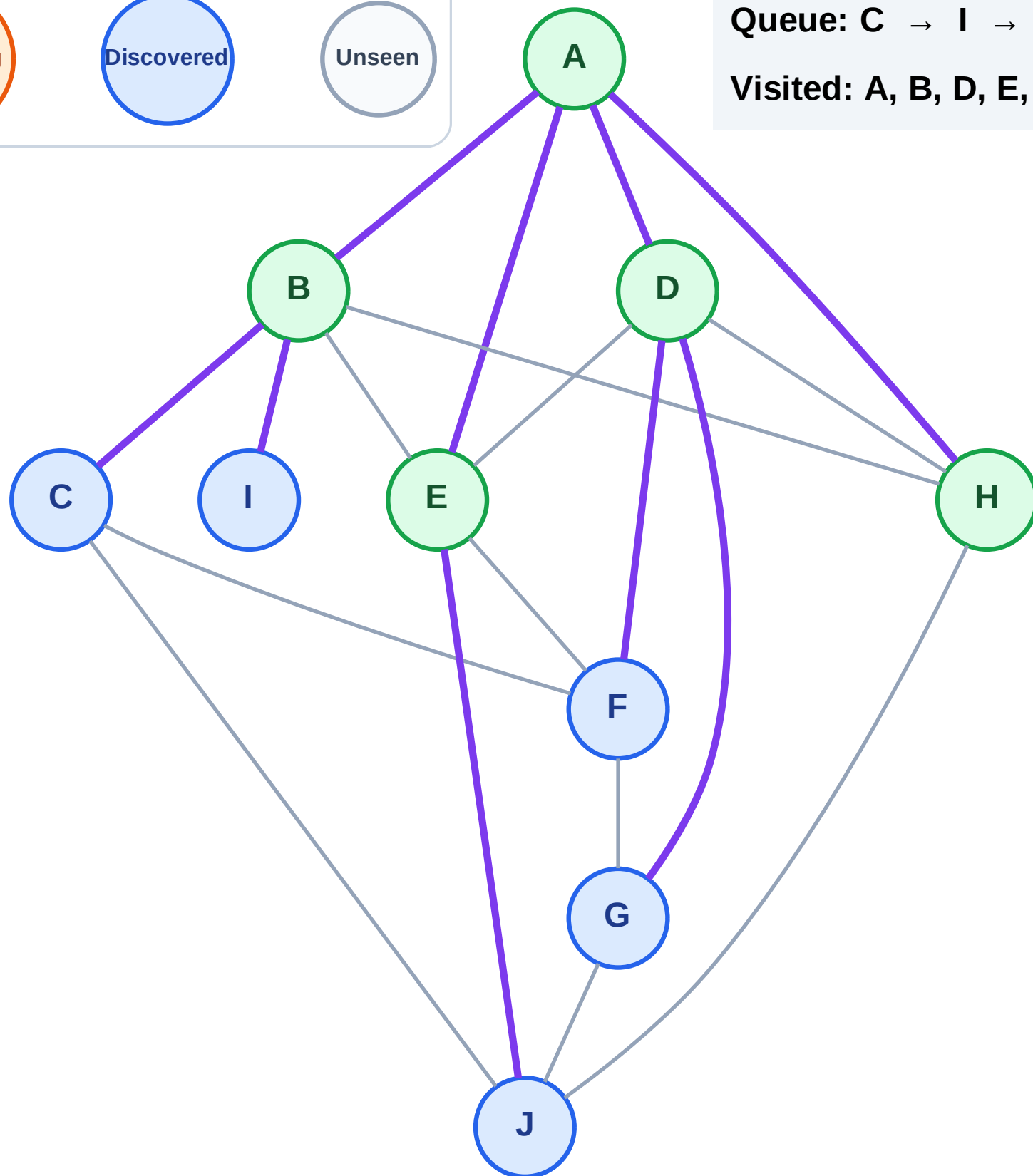
Step 11: No new neighbors from H

## Legend



Queue: C → I → F → G → J

Visited: A, B, D, E, H



# BFS Traversal

Step 12: Process node C

Legend

Complete

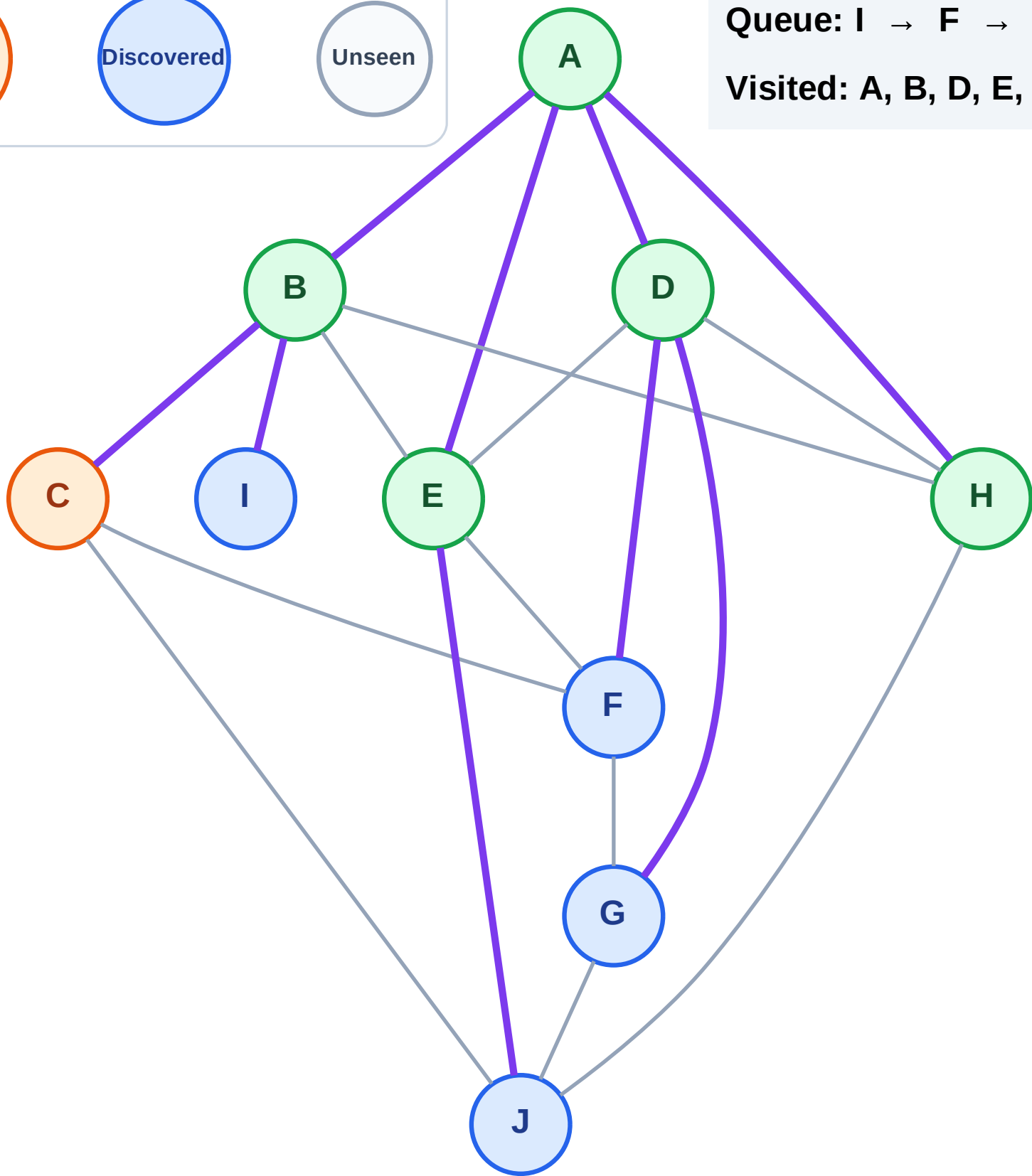
Processing

Discovered

Unseen

Queue: I → F → G → J

Visited: A, B, D, E, H



# BFS Traversal

Step 13: No new neighbors from C

Legend

Complete

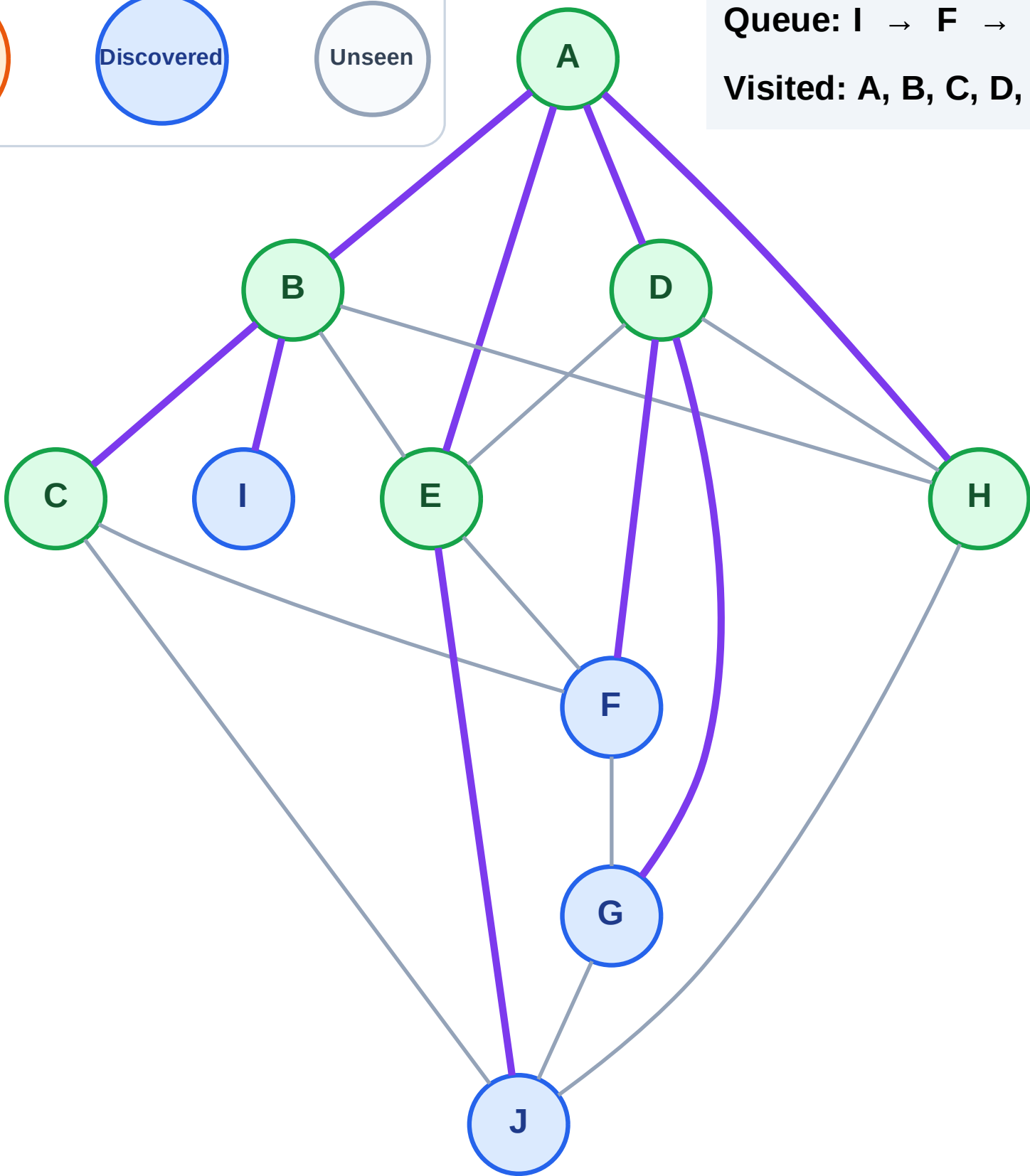
Processing

Discovered

Unseen

Queue: I → F → G → J

Visited: A, B, C, D, E, H



# BFS Traversal

Step 14: Process node I

Legend

Complete

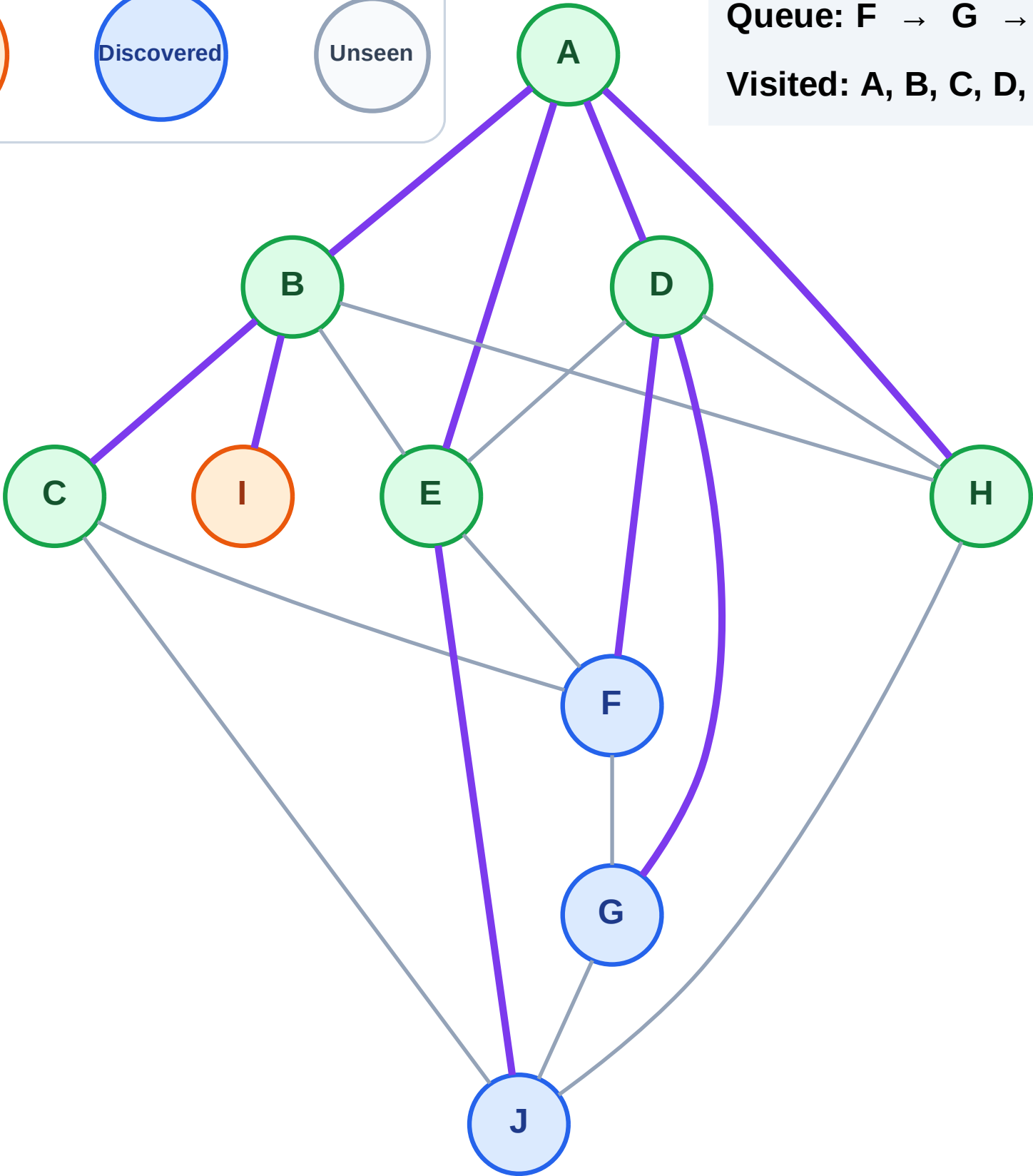
Processing

Discovered

Unseen

Queue: F → G → J

Visited: A, B, C, D, E, H



# BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

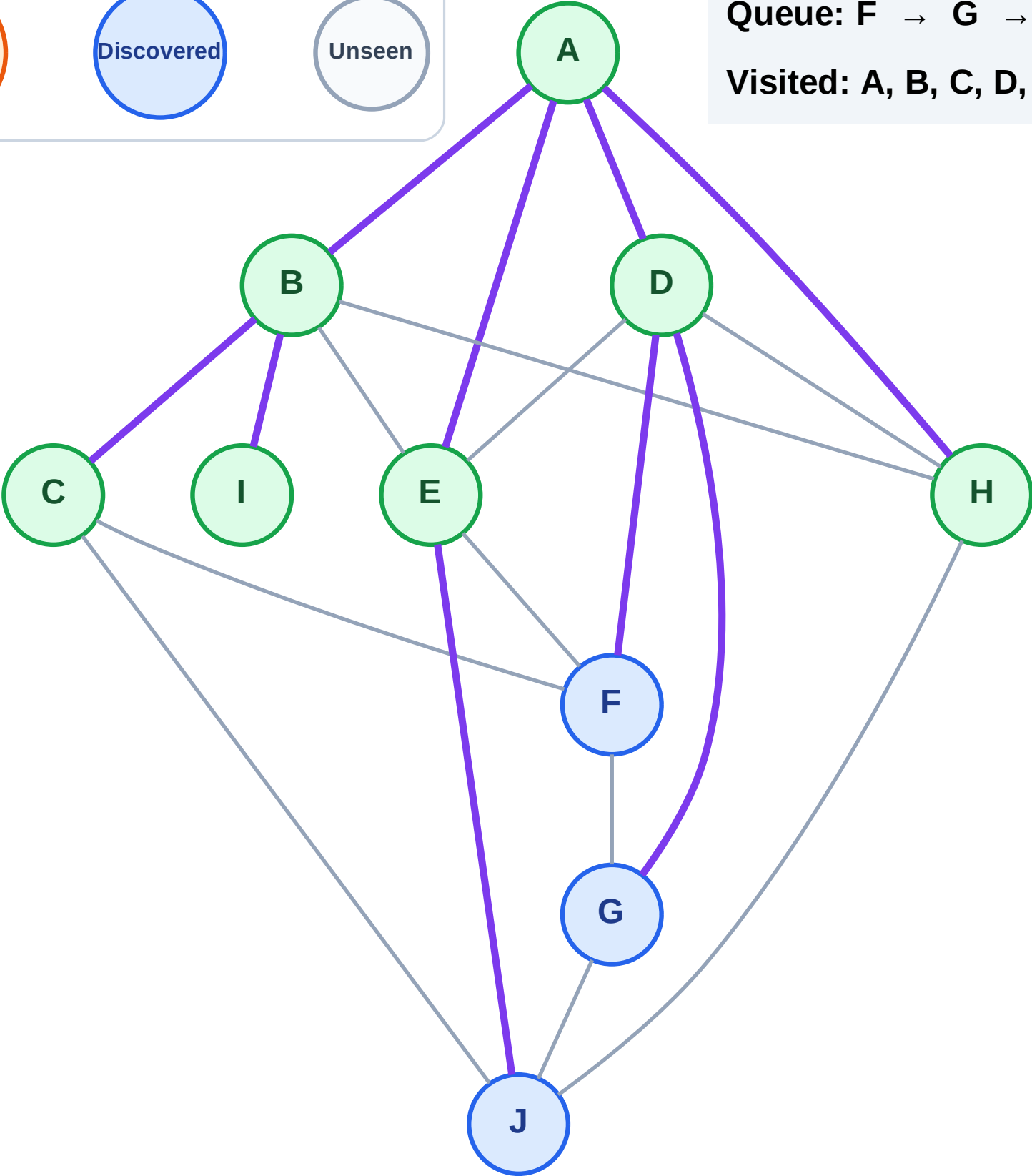
Processing

Discovered

Unseen

Queue: F → G → J

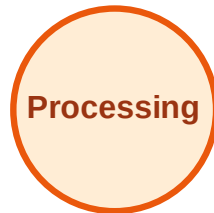
Visited: A, B, C, D, E, H, I



# BFS Traversal

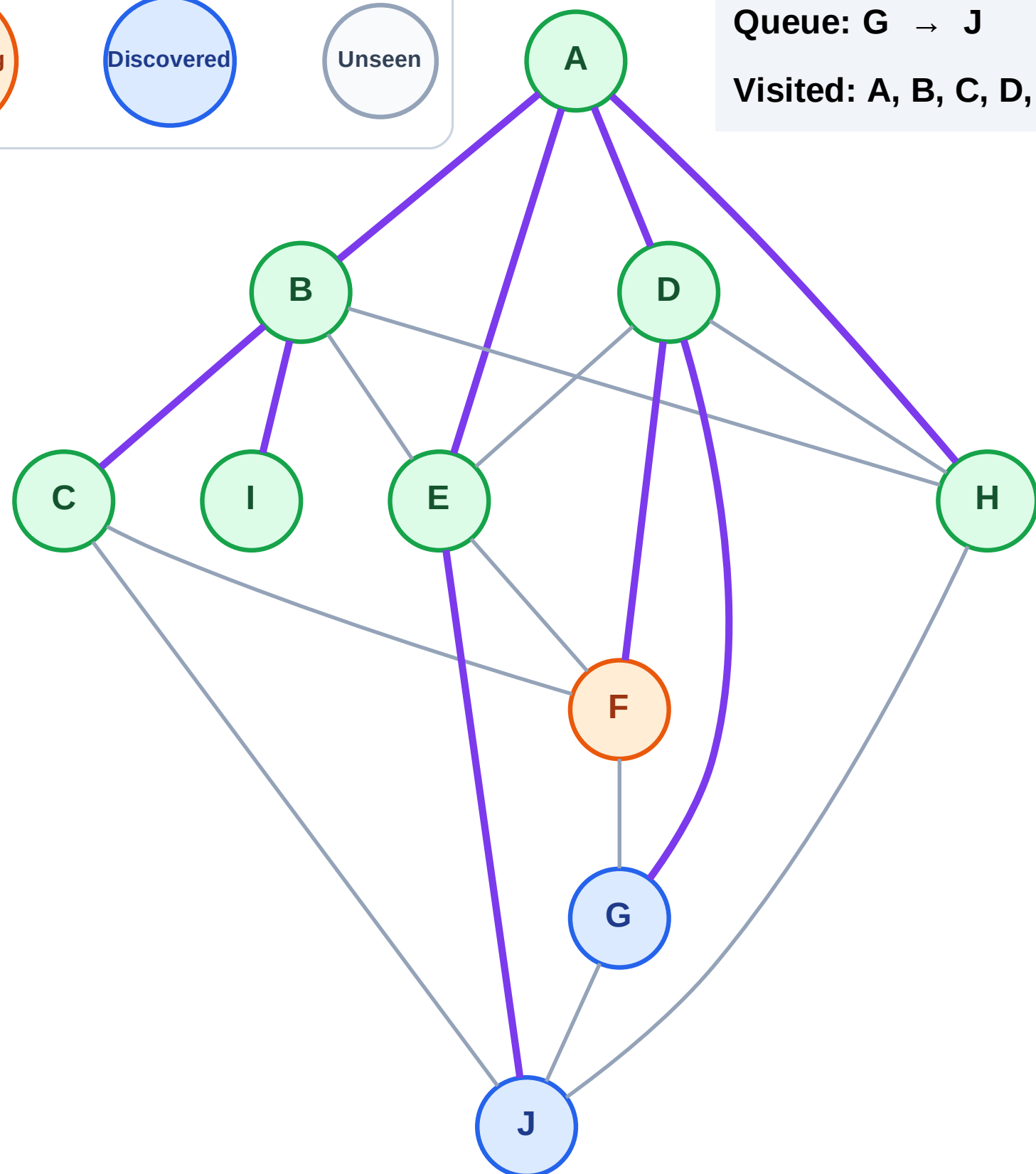
Step 16: Process node F

## Legend



Queue: G → J

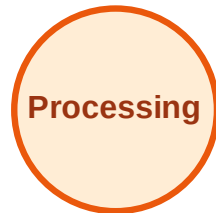
Visited: A, B, C, D, E, H, I



# BFS Traversal

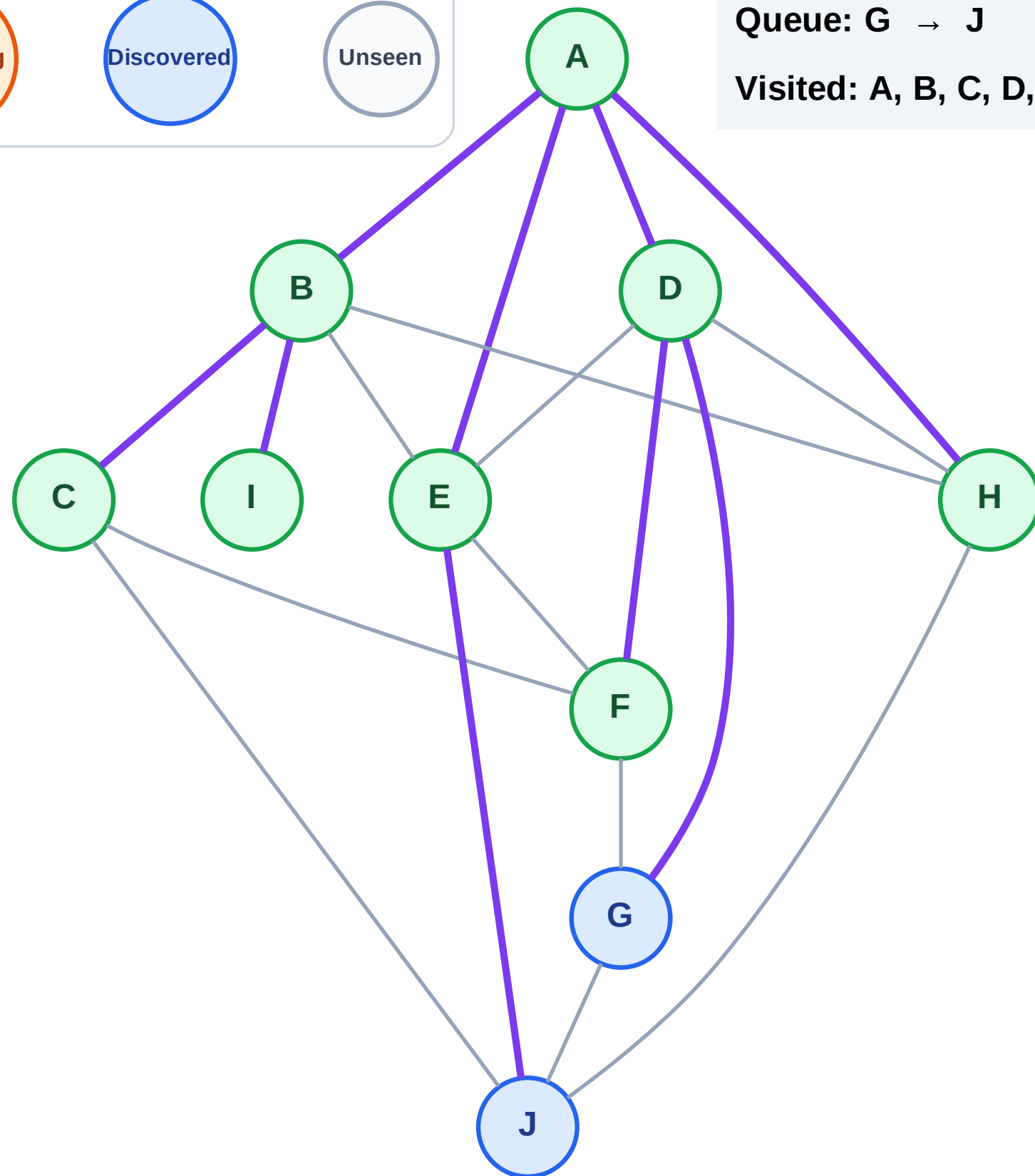
Step 17: No new neighbors from F

## Legend



Queue: G → J

Visited: A, B, C, D, E, F, H, I



# BFS Traversal

Step 18: Process node G

Legend

Complete

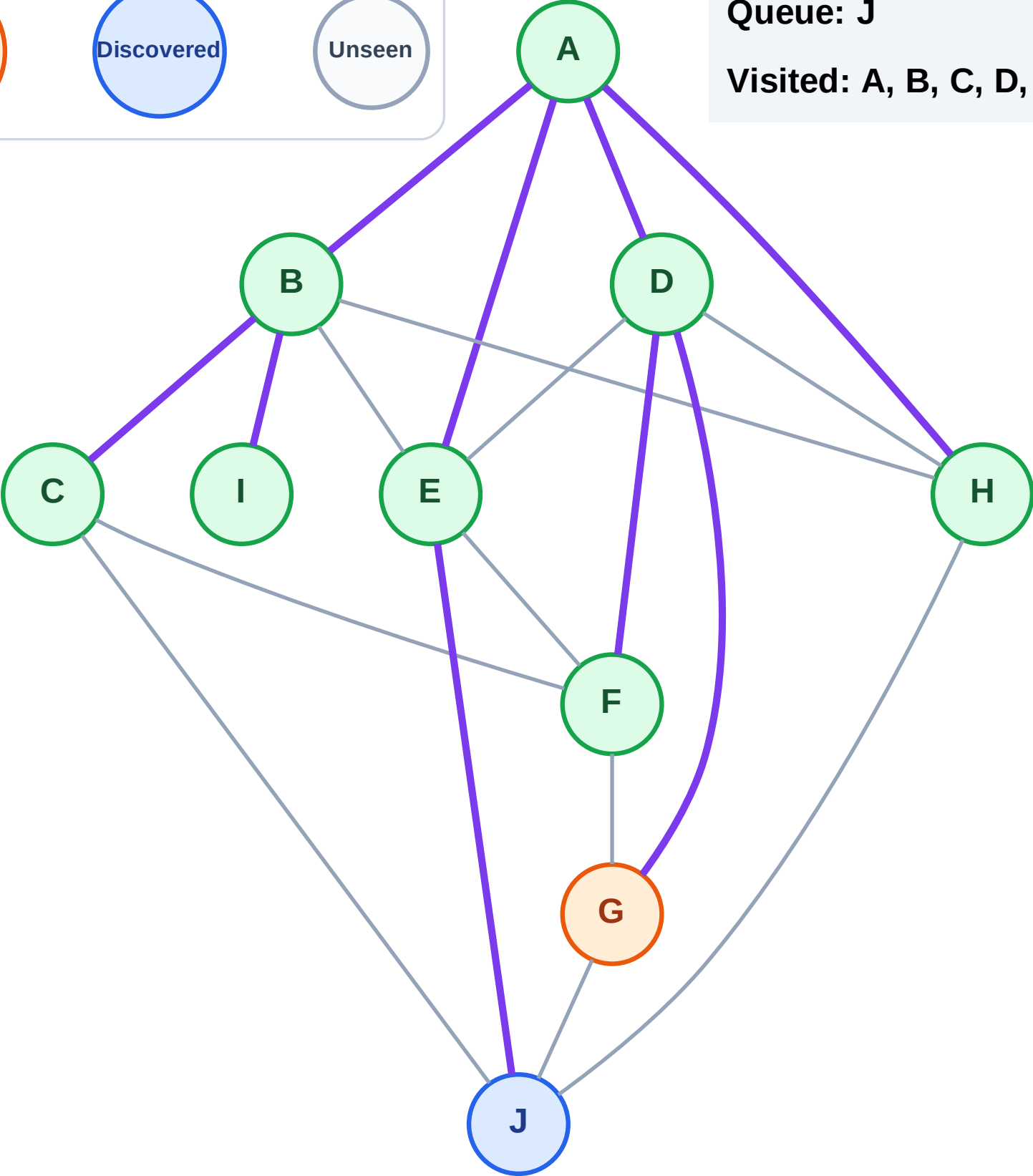
Processing

Discovered

Unseen

Queue: J

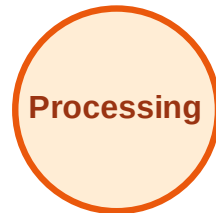
Visited: A, B, C, D, E, F, H, I



# BFS Traversal

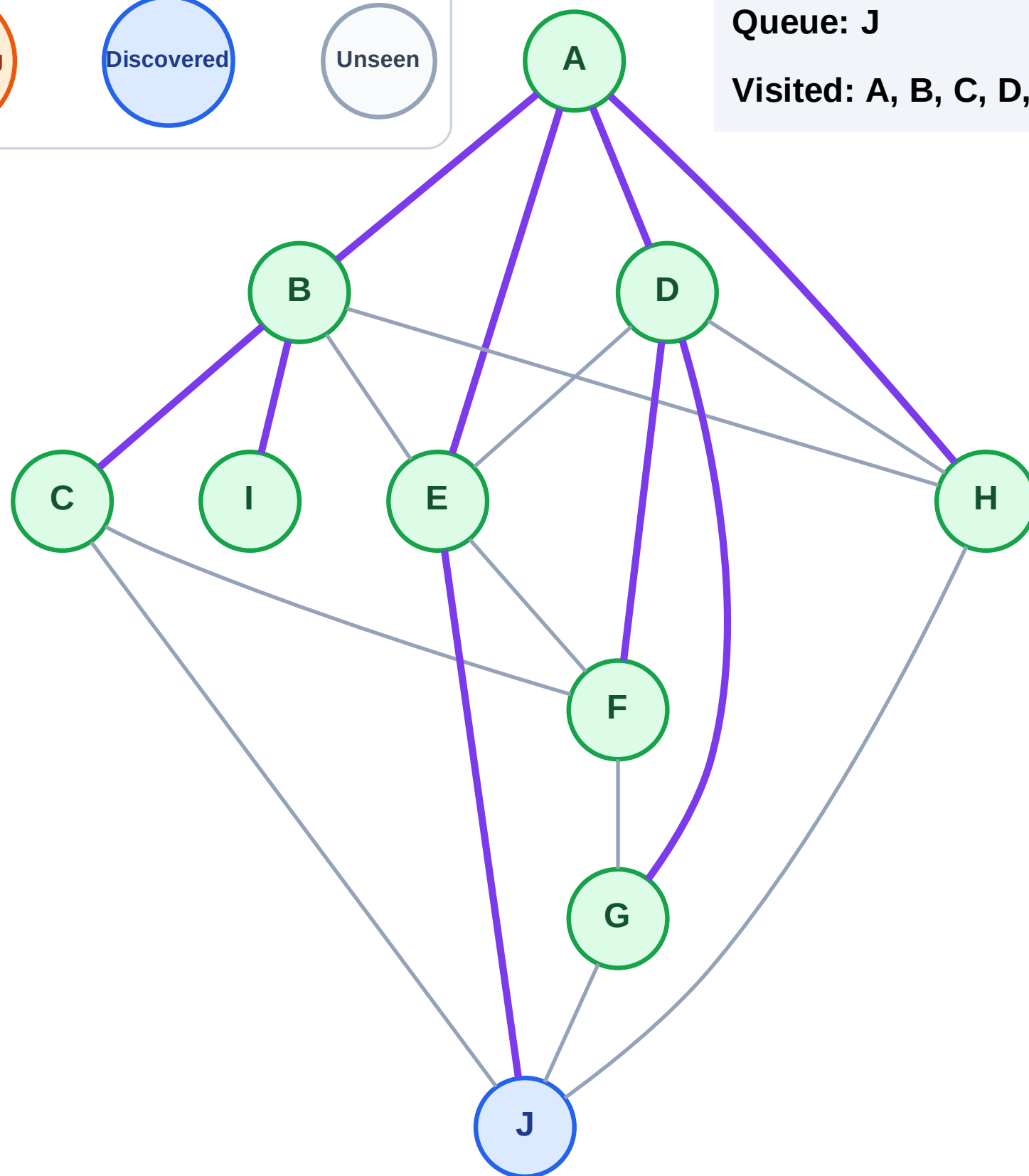
Step 19: No new neighbors from G

## Legend



Queue: J

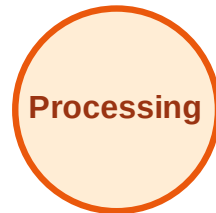
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

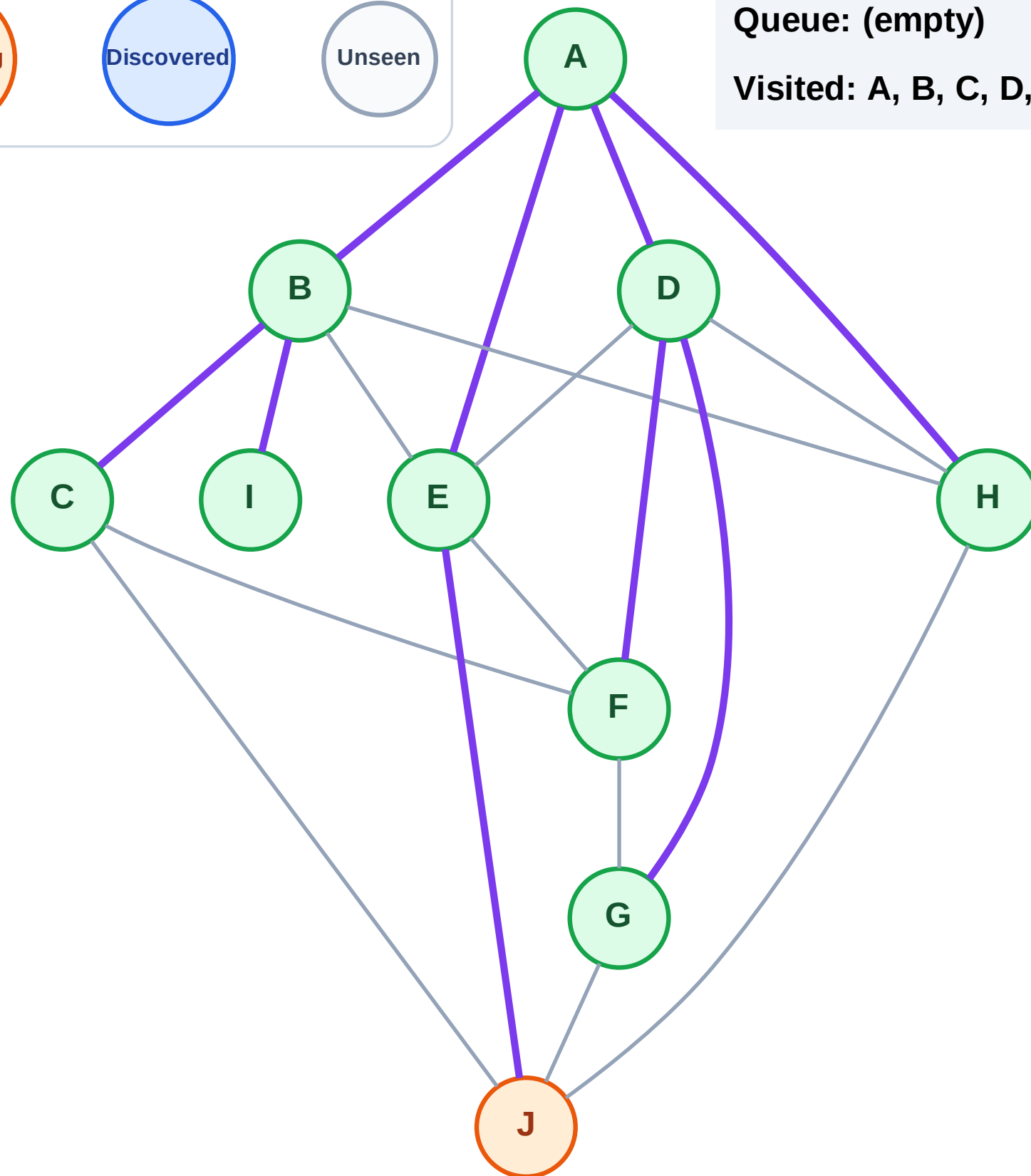
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

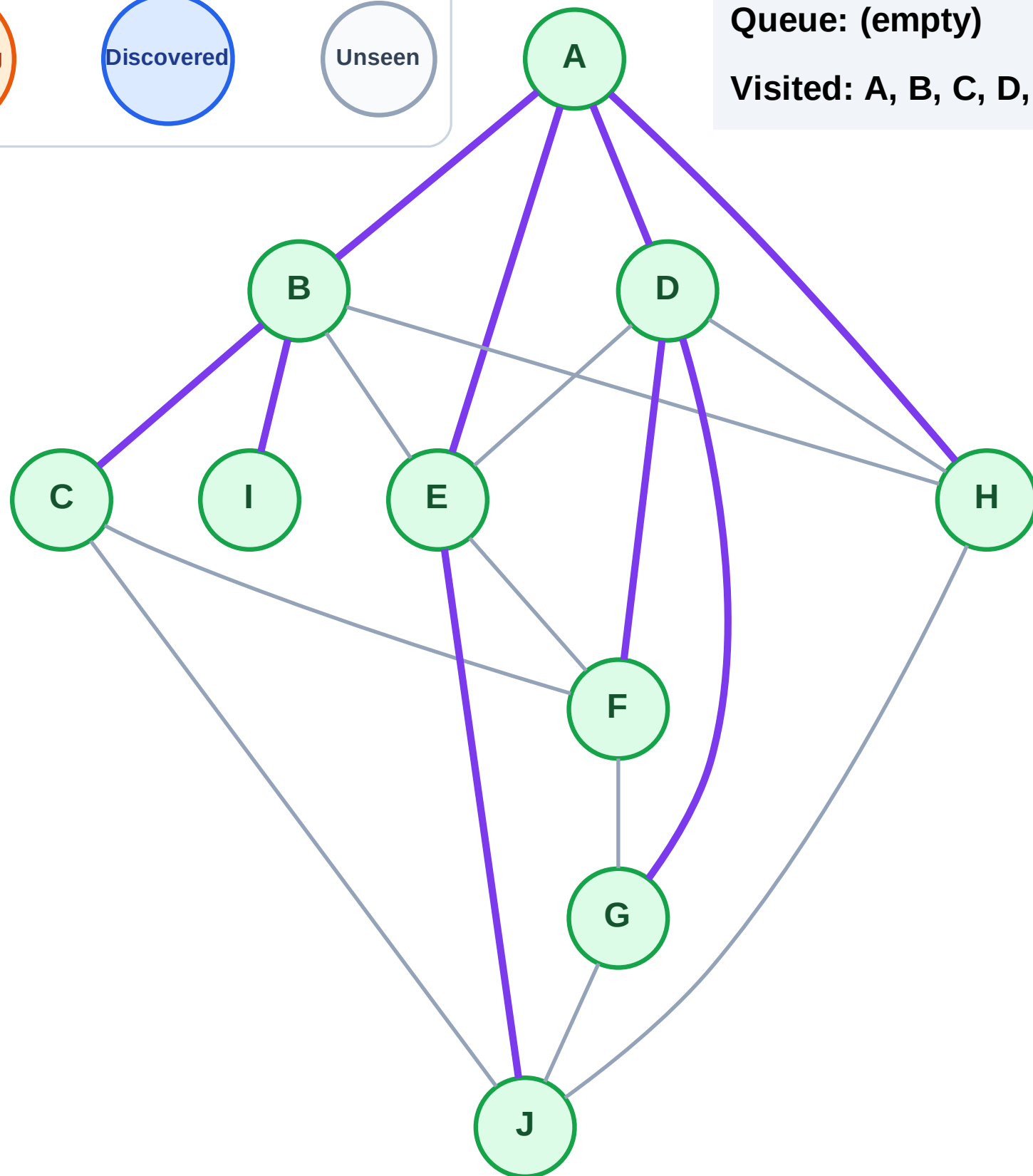
Step 21: No new neighbors from J

## Legend



Queue: (empty)

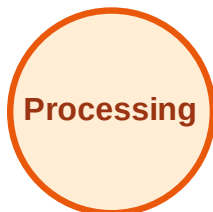
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

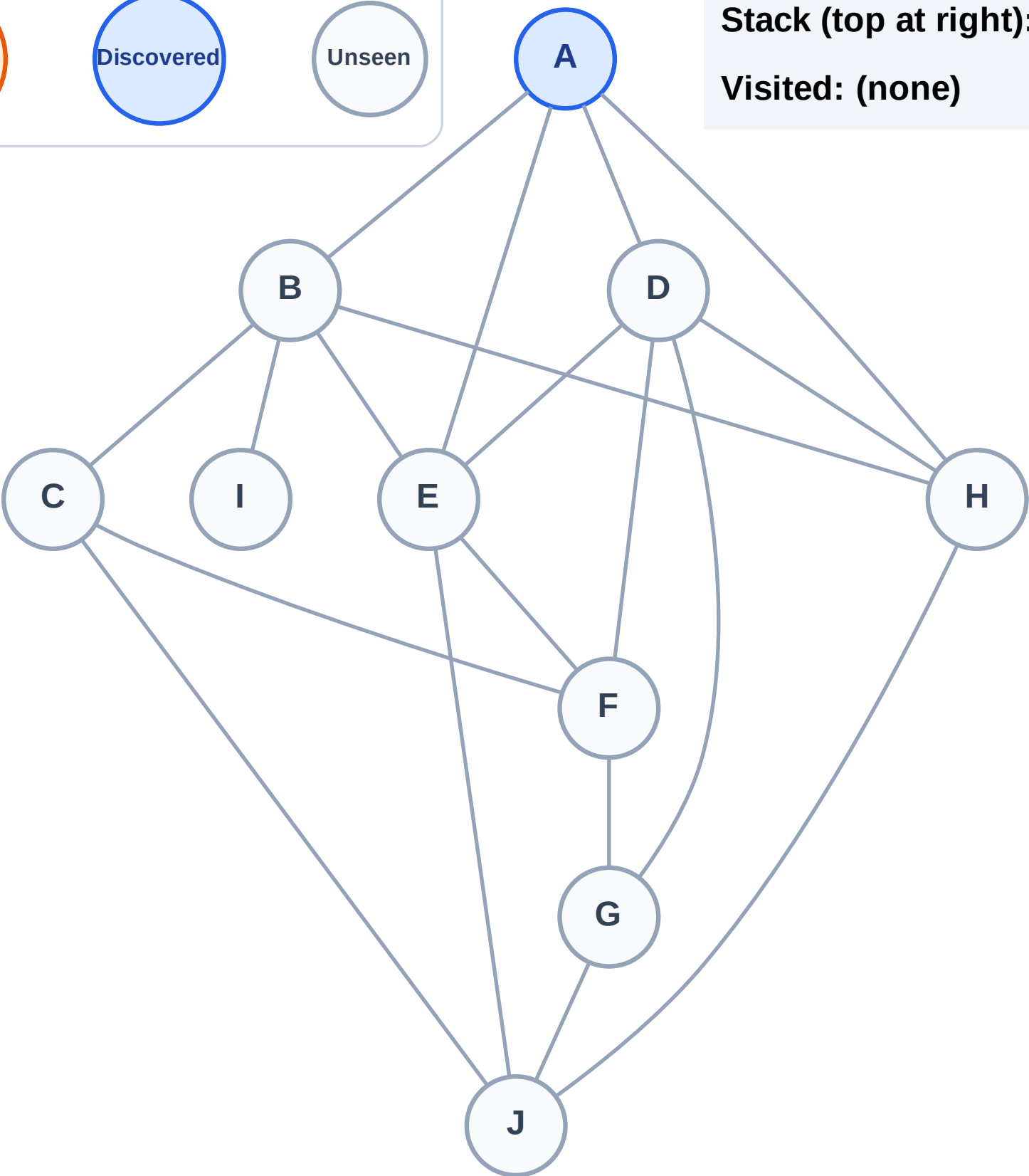
Step 1: Start at A

## Legend



Stack (top at right): A

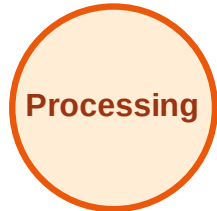
Visited: (none)



# DFS Traversal

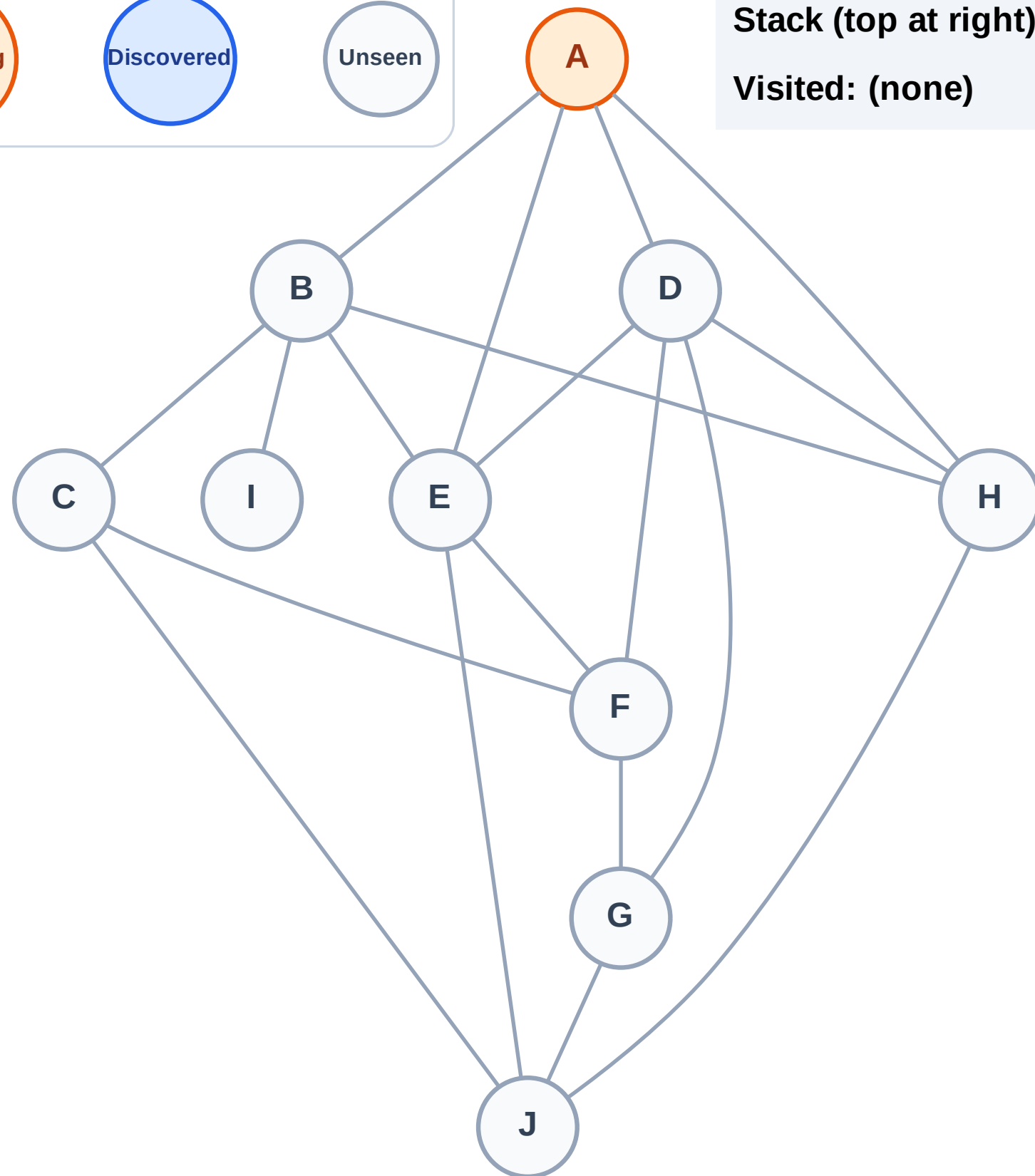
Step 2: Process node A

## Legend



Stack (top at right): (empty)

Visited: (none)



# DFS Traversal

Step 3: Discover H, E, D, B from A

Legend

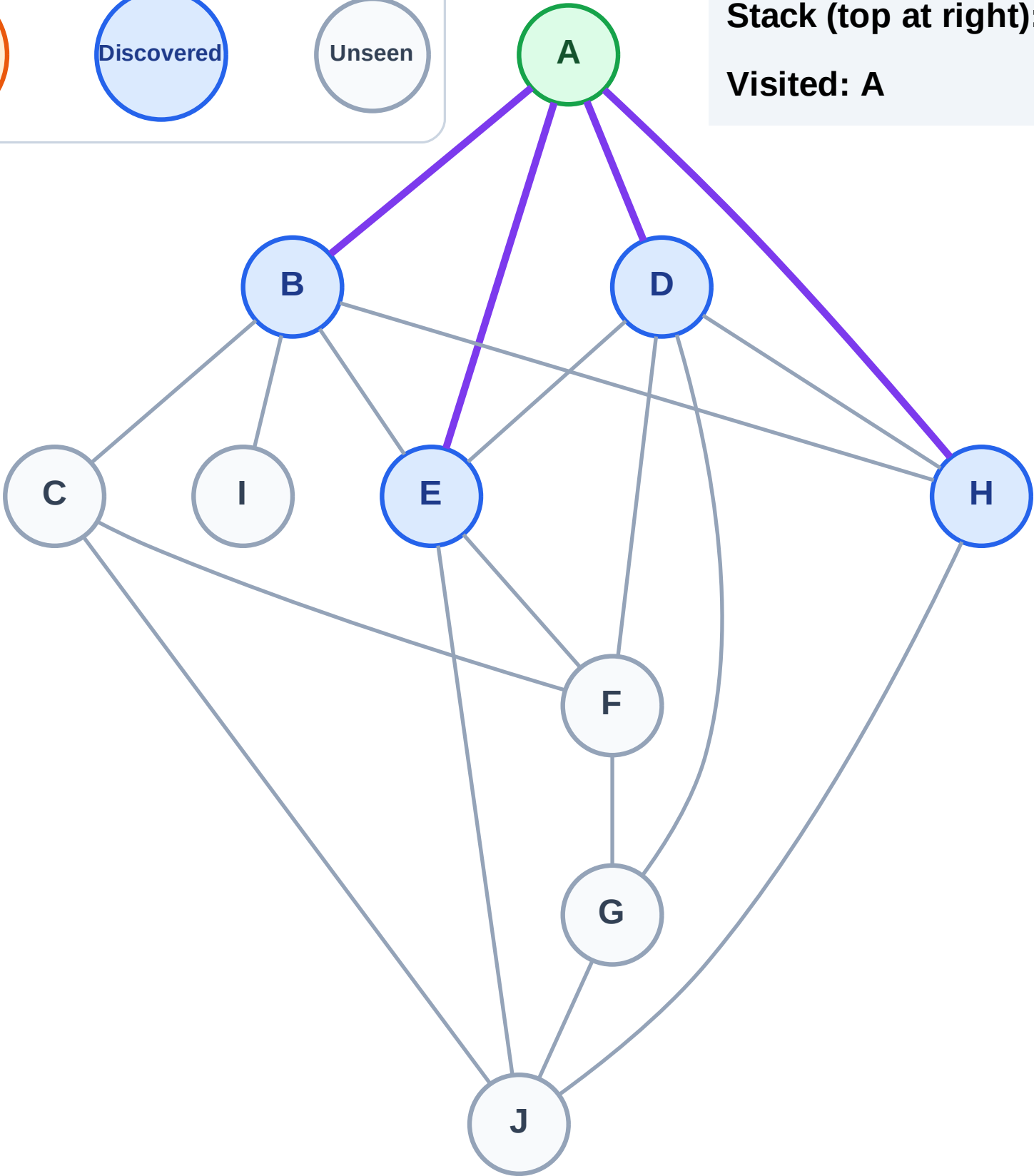
Complete

Processing

Discovered

Unseen

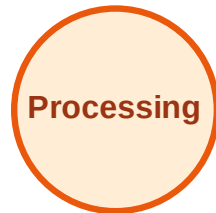
Stack (top at right): H | E | D | B  
Visited: A



# DFS Traversal

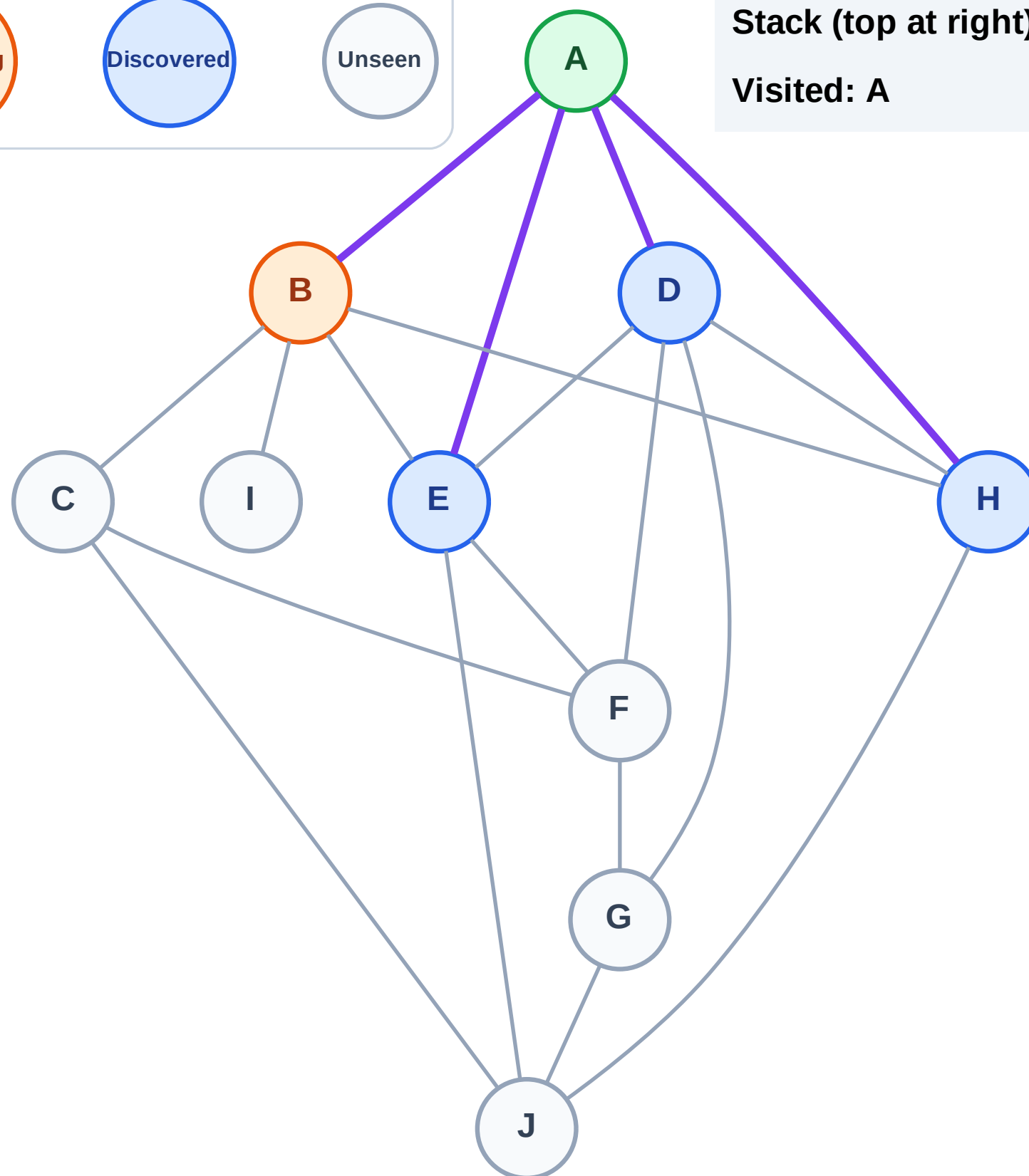
Step 4: Process node B

## Legend



Stack (top at right): H | E | D

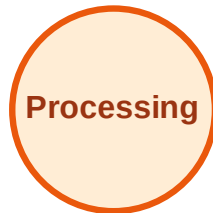
Visited: A



# DFS Traversal

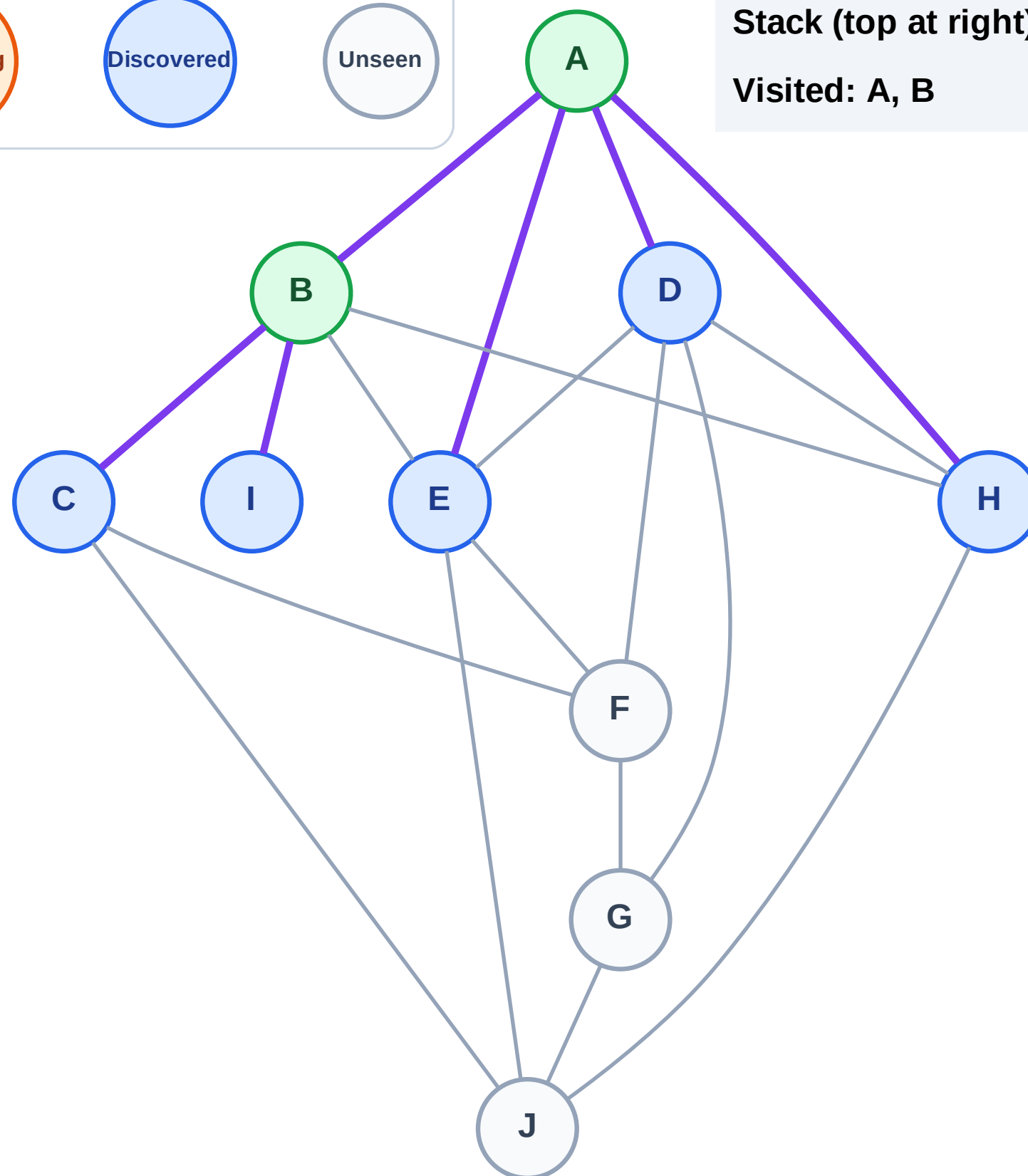
Step 5: Discover I, C from B

## Legend



Stack (top at right): H | E | D | I | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

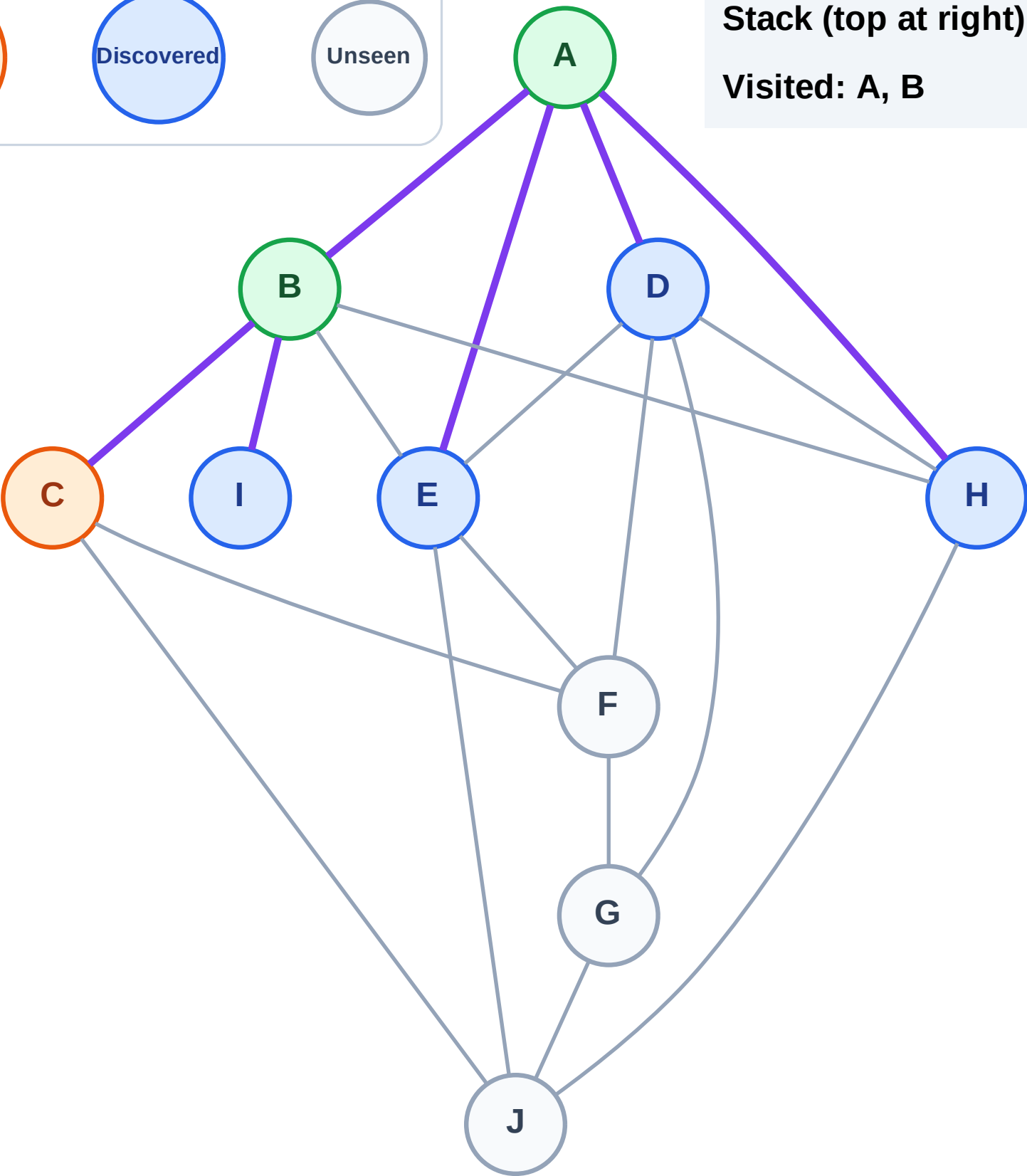
Complete

Processing

Discovered

Unseen

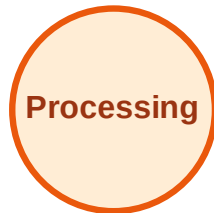
Stack (top at right): H | E | D | I  
Visited: A, B



# DFS Traversal

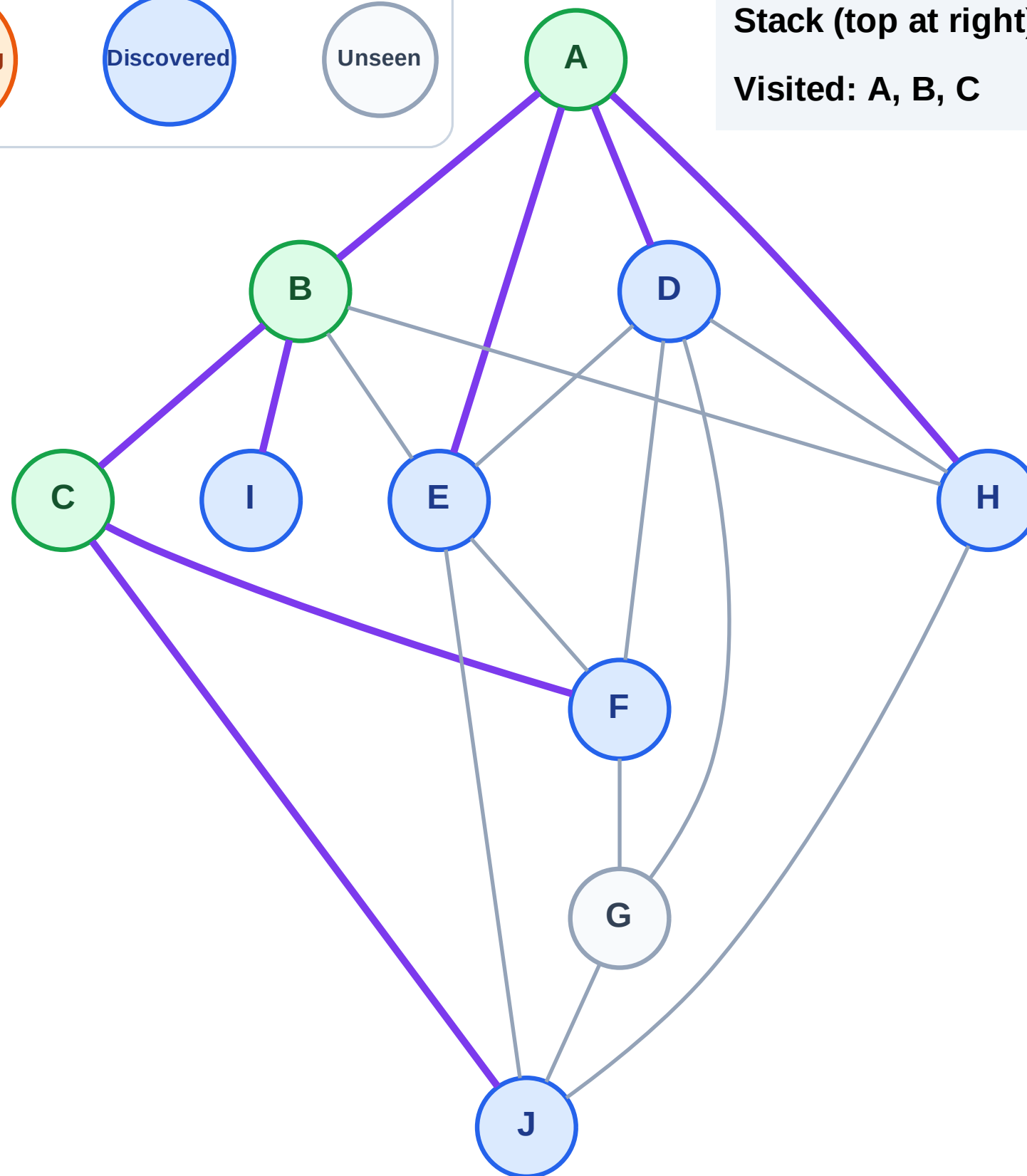
Step 7: Discover J, F from C

## Legend



Stack (top at right): H | E | D | I | J | F

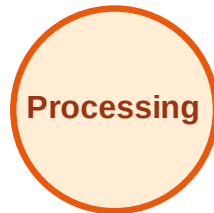
Visited: A, B, C



# DFS Traversal

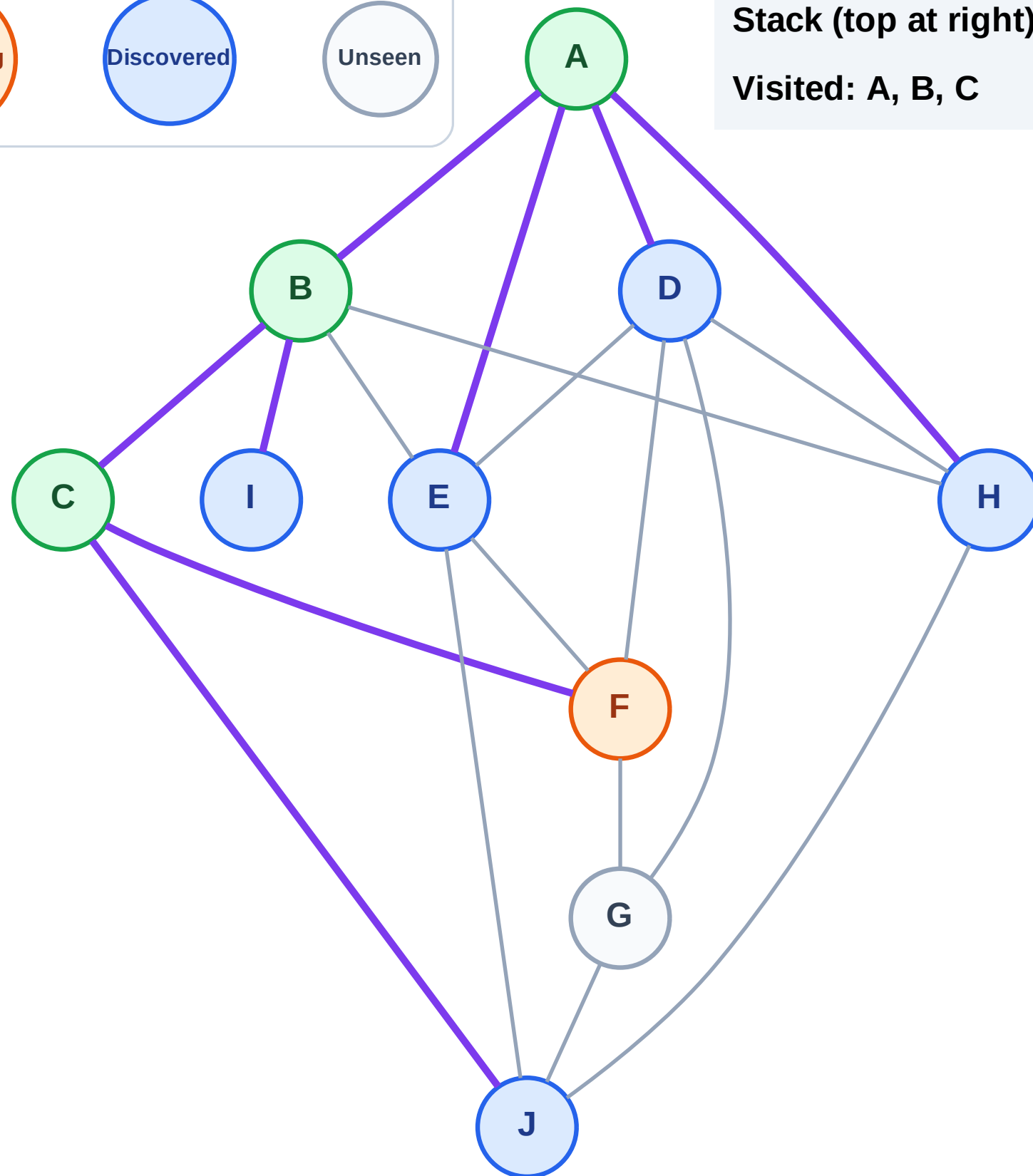
Step 8: Process node F

## Legend



Stack (top at right): H | E | D | I | J

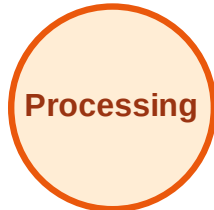
Visited: A, B, C



# DFS Traversal

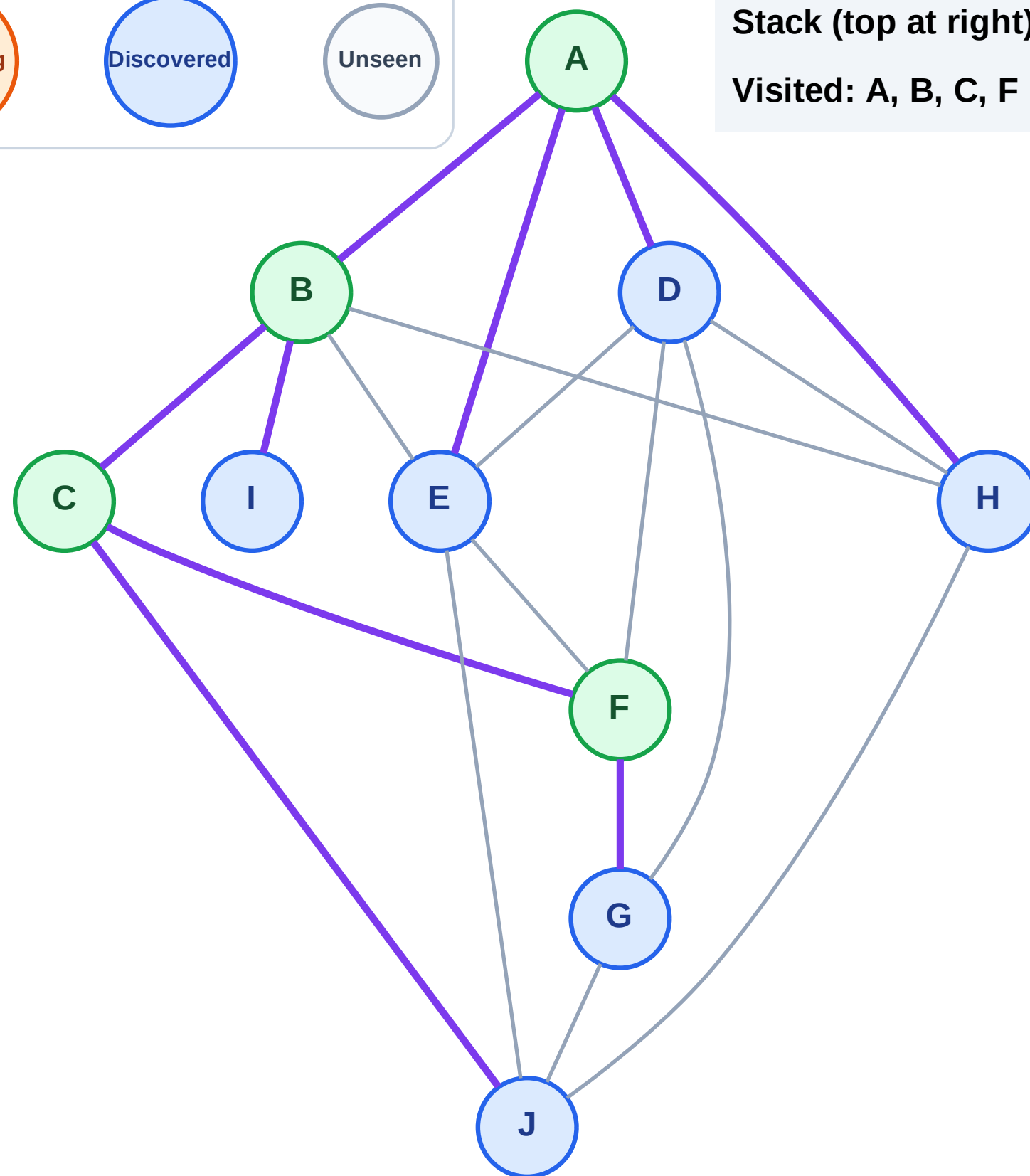
Step 9: Discover G from F

## Legend



Stack (top at right): H | E | D | I | J | G

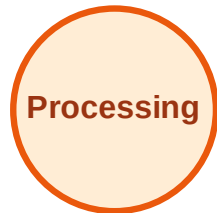
Visited: A, B, C, F



# DFS Traversal

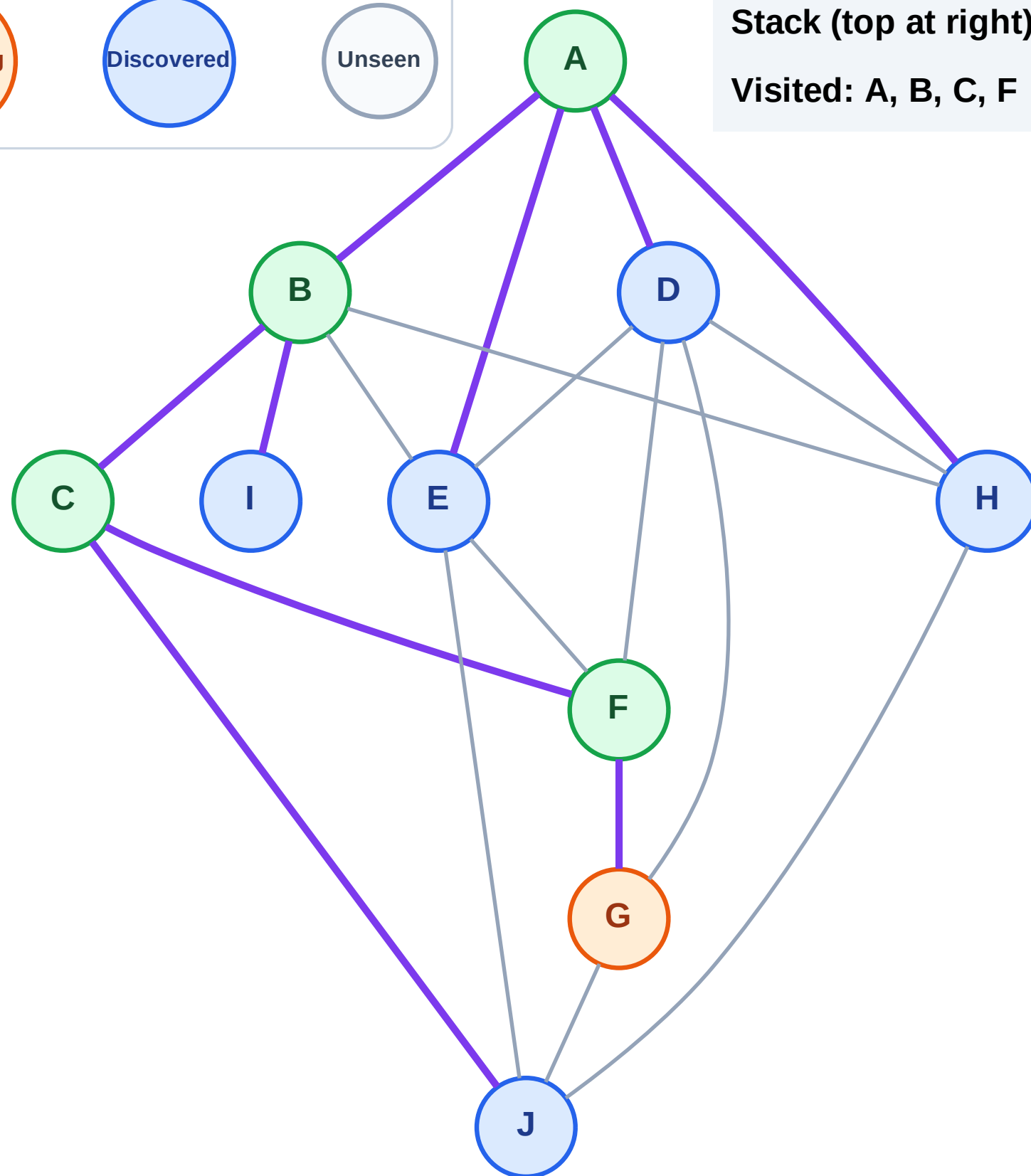
Step 10: Process node G

## Legend



Stack (top at right): H | E | D | I | J

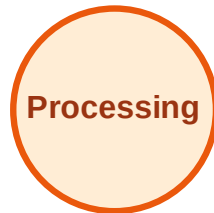
Visited: A, B, C, F



# DFS Traversal

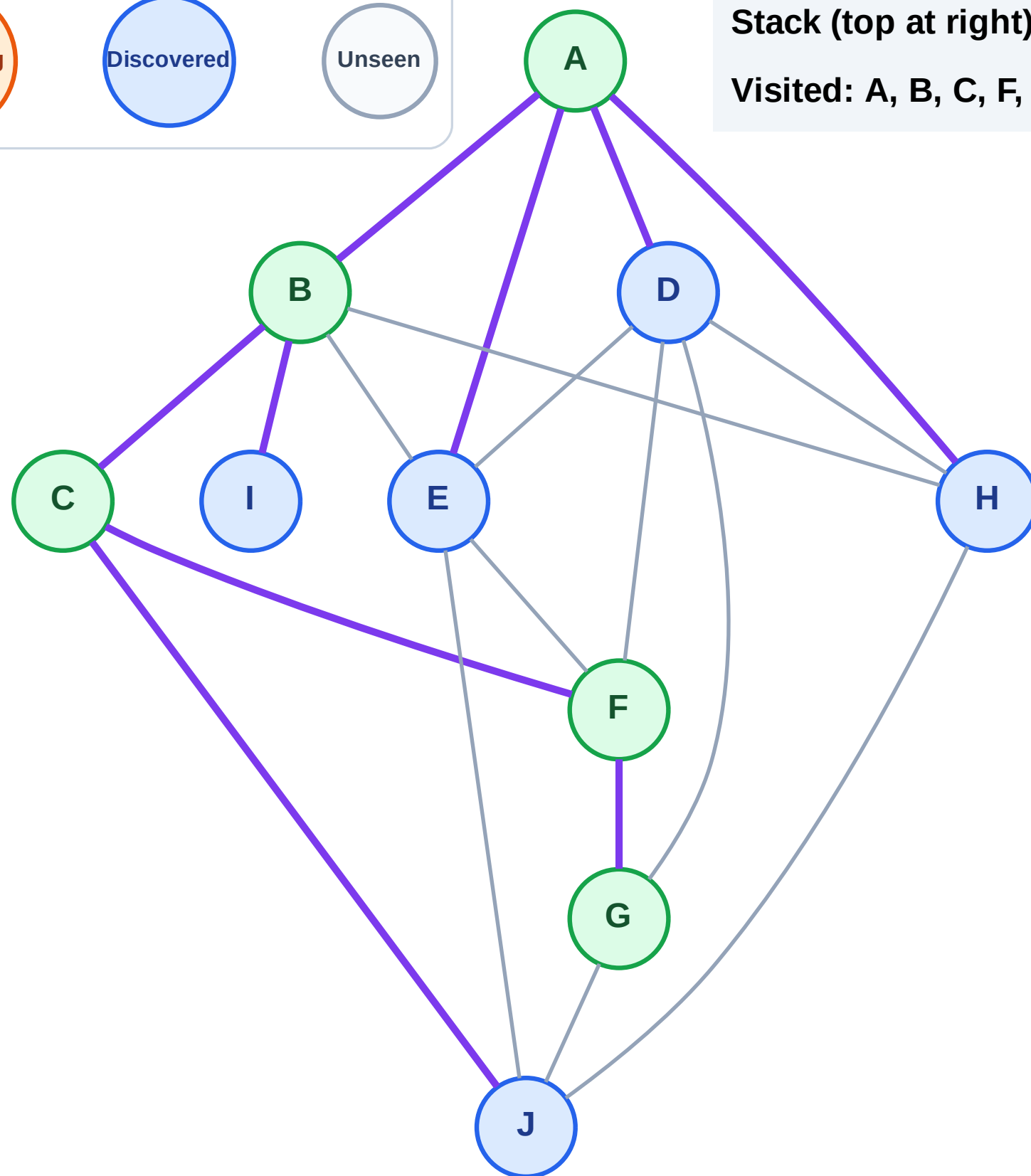
Step 11: No new neighbors from G

## Legend



Stack (top at right): H | E | D | I | J

Visited: A, B, C, F, G



# DFS Traversal

Step 12: Process node J

Legend

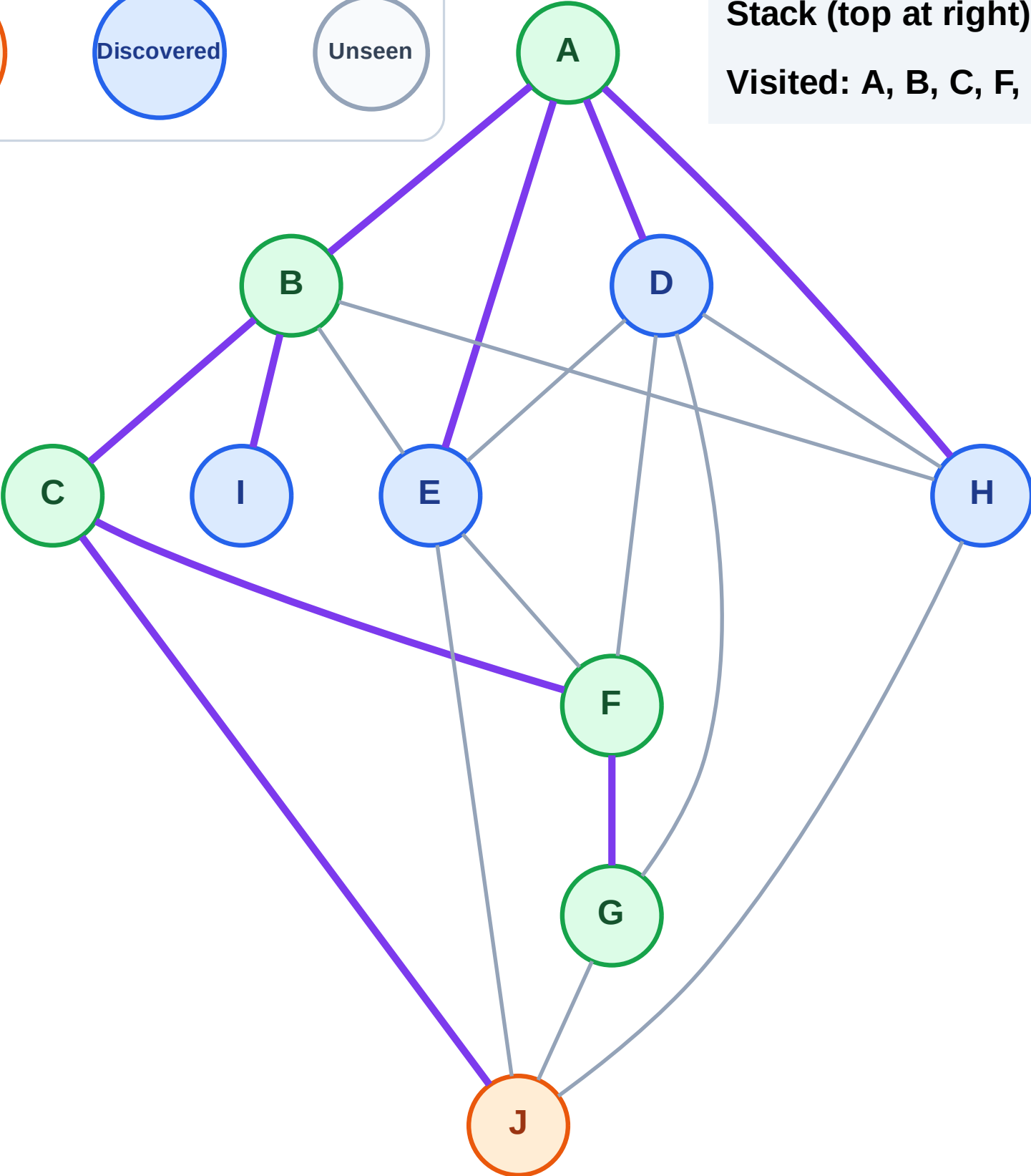
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | D | I  
Visited: A, B, C, F, G



# DFS Traversal

Step 13: No new neighbors from J

Legend

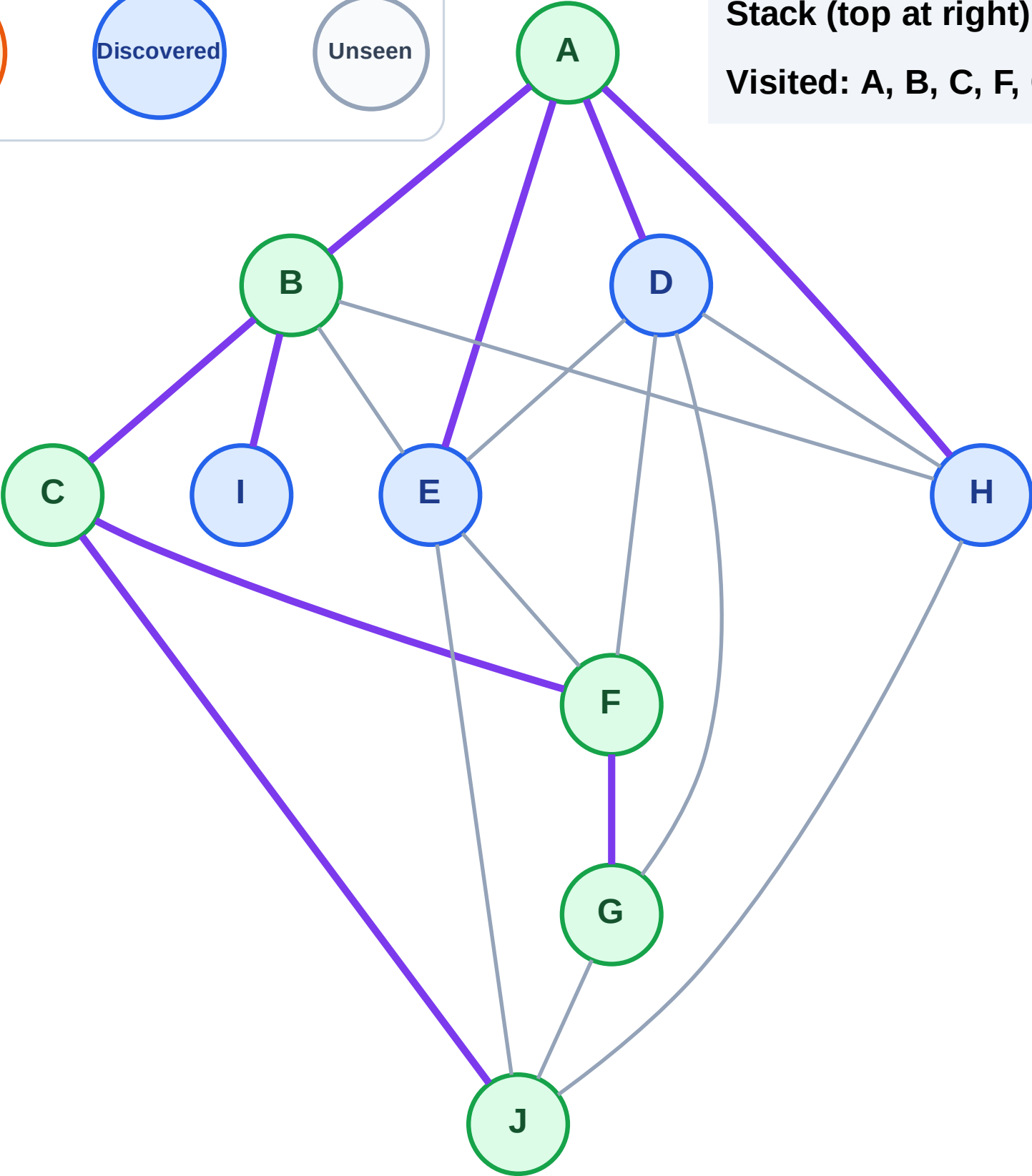
Complete

Processing

Discovered

Unseen

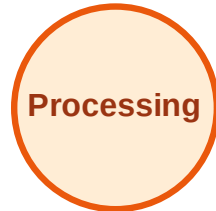
Stack (top at right): H | E | D | I  
Visited: A, B, C, F, G, J



# DFS Traversal

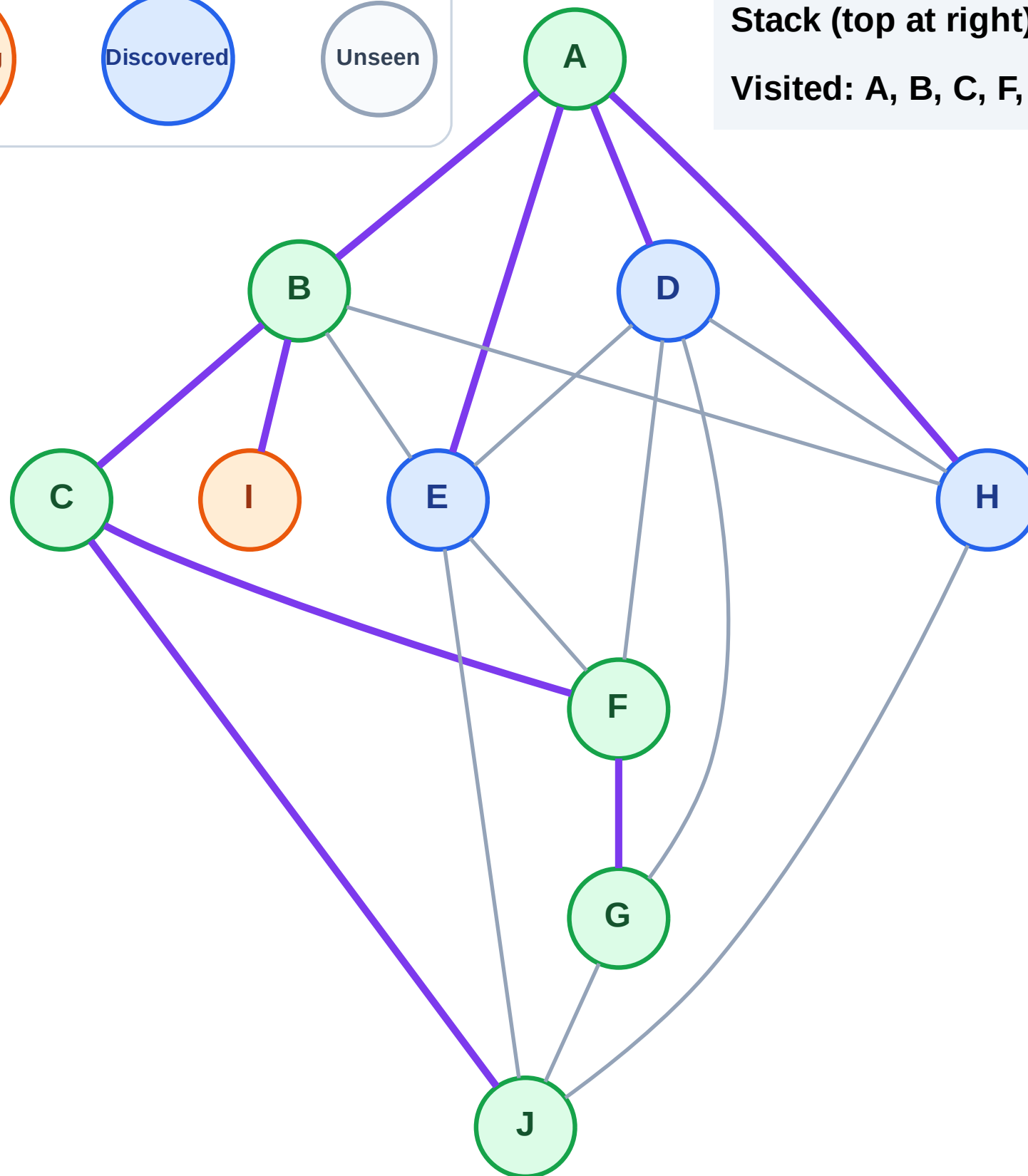
Step 14: Process node I

## Legend



Stack (top at right): H | E | D

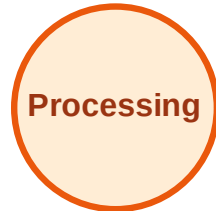
Visited: A, B, C, F, G, J



# DFS Traversal

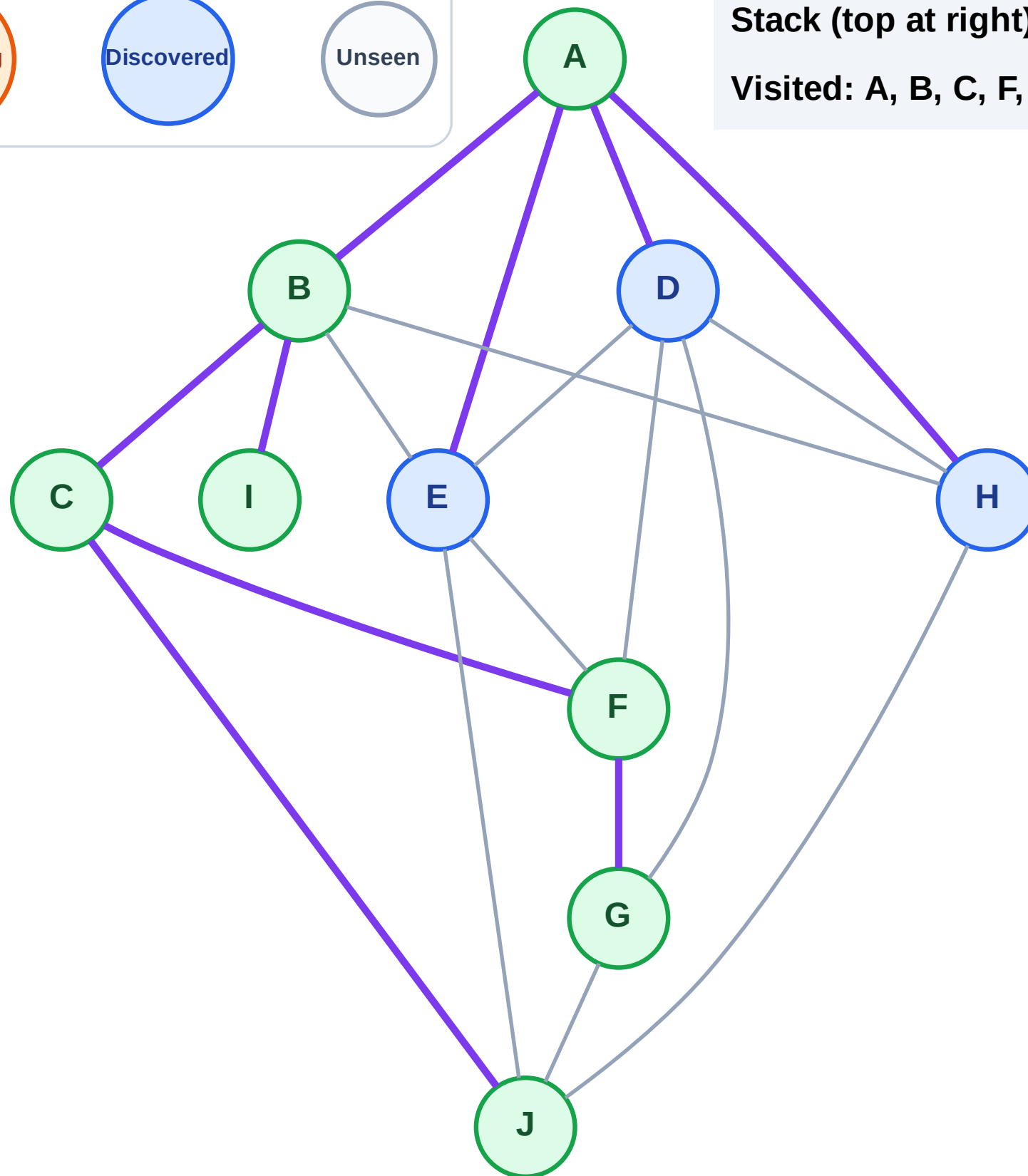
Step 15: No new neighbors from I

## Legend



Stack (top at right): H | E | D

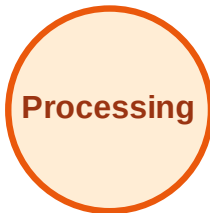
Visited: A, B, C, F, G, I, J



# DFS Traversal

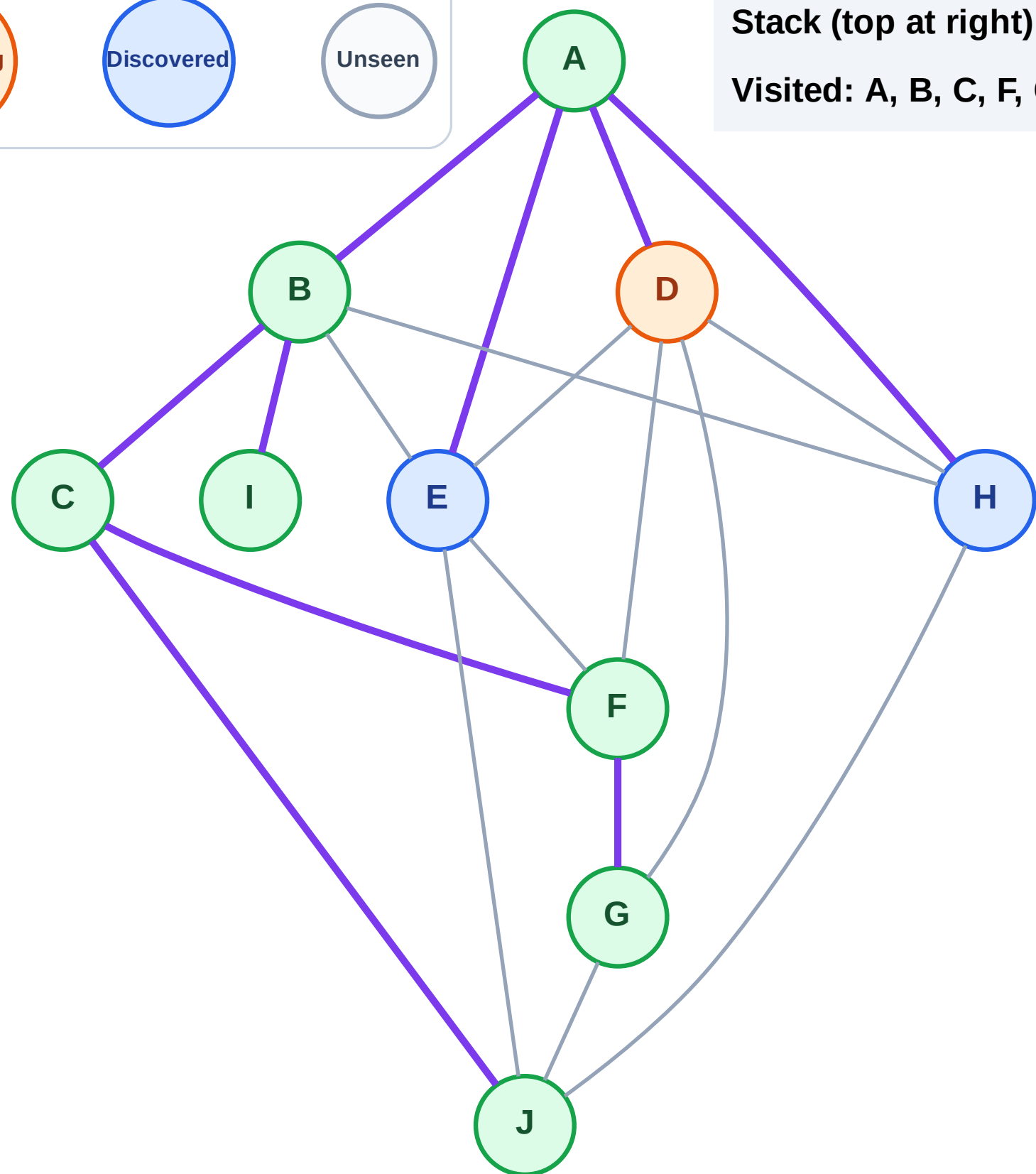
Step 16: Process node D

## Legend



Stack (top at right): H | E

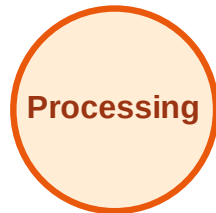
Visited: A, B, C, F, G, I, J



# DFS Traversal

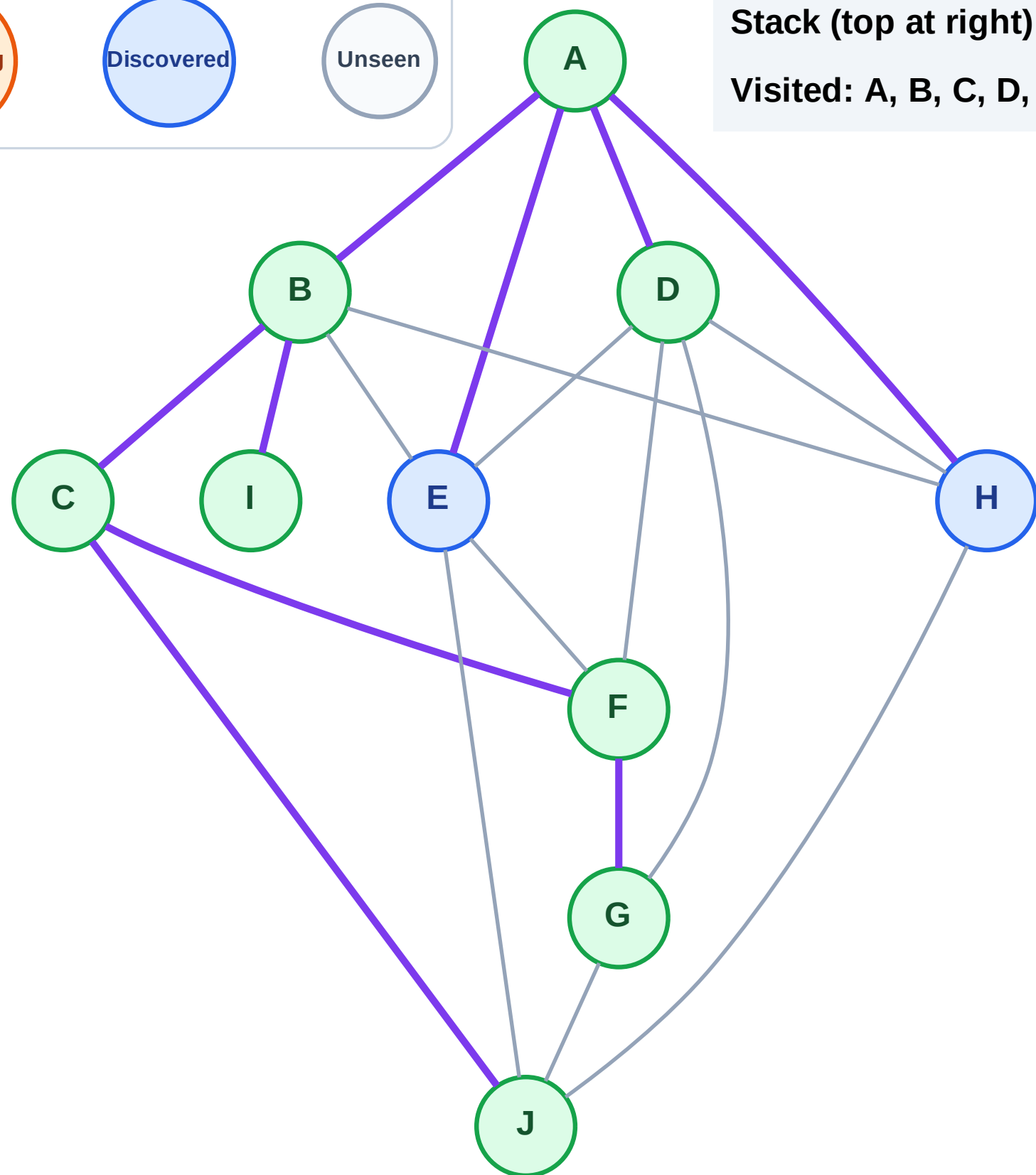
Step 17: No new neighbors from D

## Legend



Stack (top at right): H | E

Visited: A, B, C, D, F, G, I, J



# DFS Traversal

Step 18: Process node E

Legend

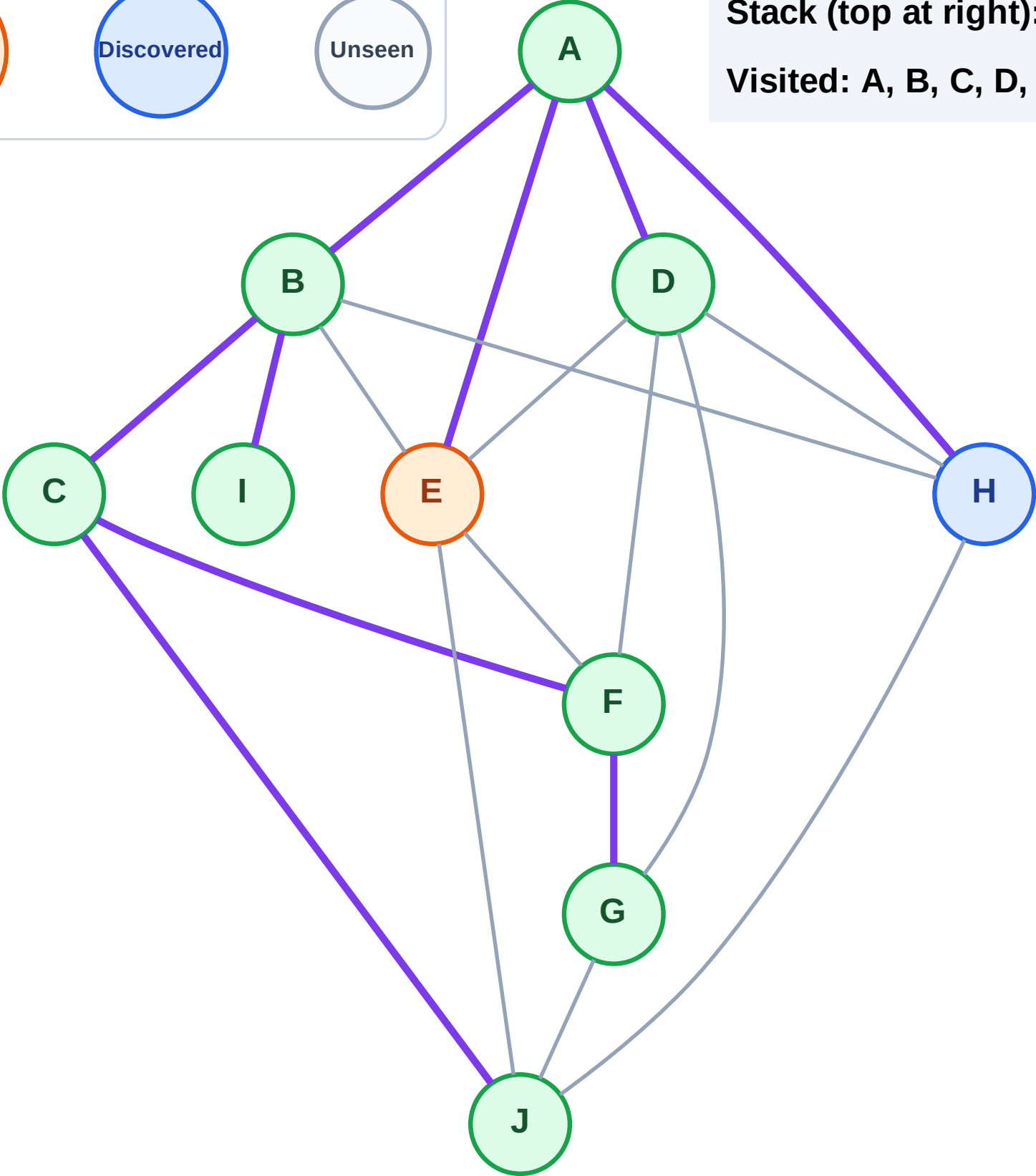
Complete

Processing

Discovered

Unseen

Stack (top at right): H  
Visited: A, B, C, D, F, G, I, J

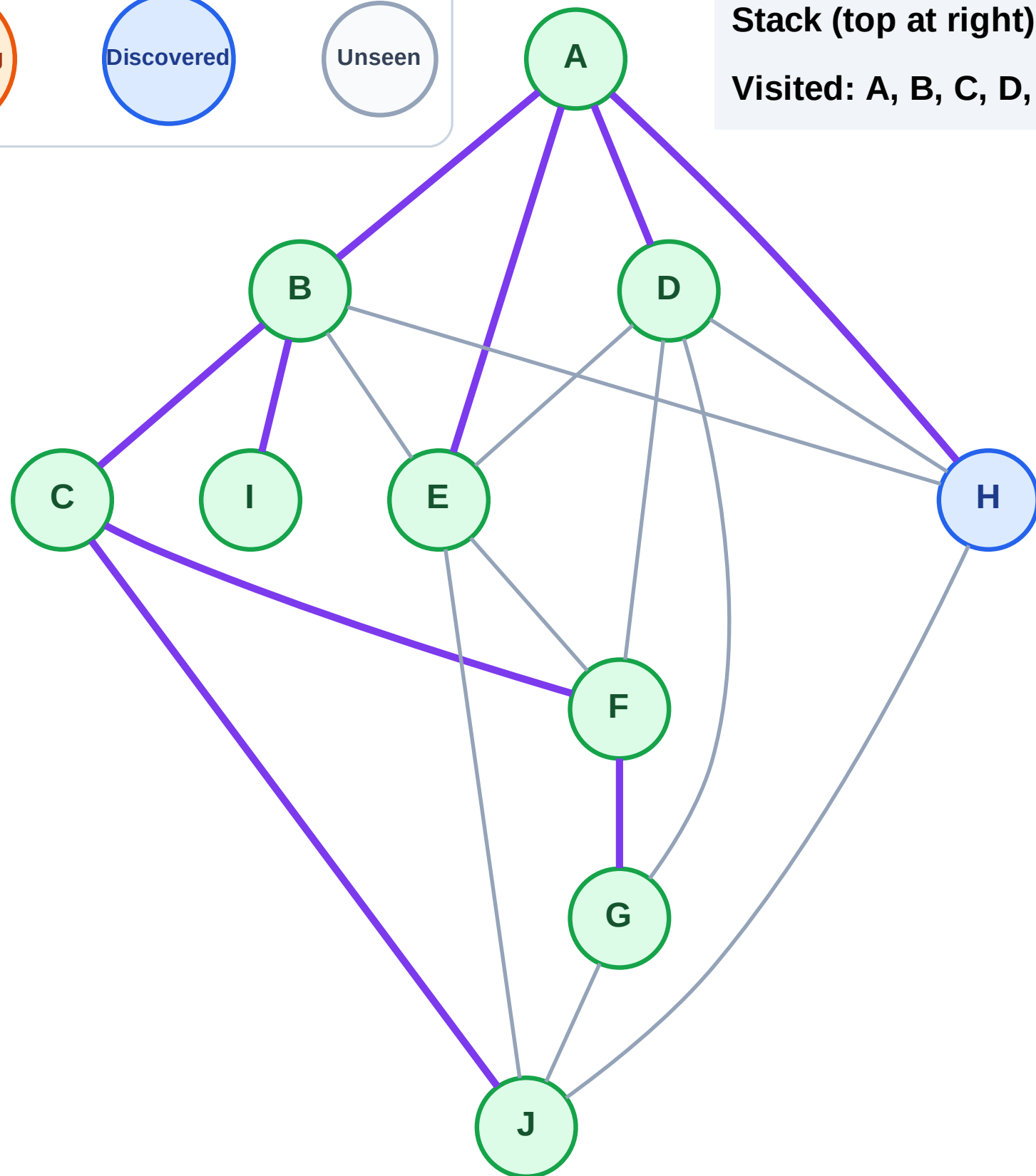


Step 19: No new neighbors from E

### Legend

## Unseen

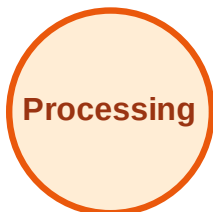
**Visited: A, B, C, D, E, F, G, I, J**



# DFS Traversal

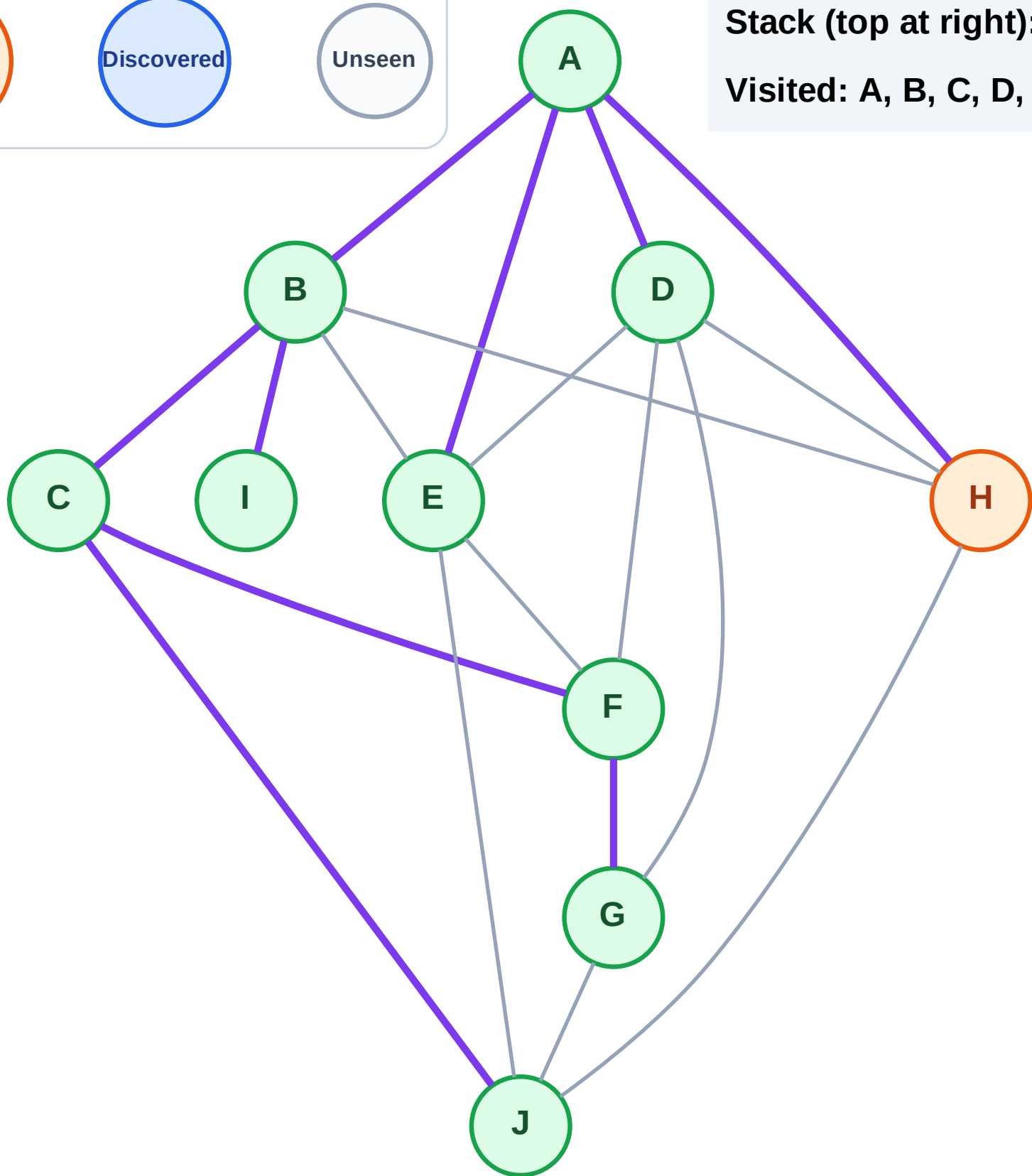
Step 20: Process node H

## Legend



Stack (top at right): (empty)

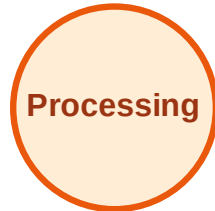
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

